

REVIEW

Optimisation methods for dispatch and control of energy storage with renewable integration

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Abstract

Renewable energy integration is an effective measure to resolve environmental problems and implement sustainable development, yet the volatility of wind and solar generation has a profound impact on power system operation, calling for sufficient backup capacity. Energy storage can shift demand over time and mitigate real-time power mismatch and thus help integrate renewable energy resources into power grids. However, the unit capacity price of energy storage is still relatively high, and the capacity of energy storage is usually limited. Given the prominent uncertainty and finite capacity of energy storage, it is crucially important to take full advantage of energy storage units by strategic dispatch and control. From the mathematical point of view, energy storage dispatch and control give rise to a sequential decision-making process involving uncertain parameters and inter-temporal constraints. Multitudinous optimisation methods have been developed for such a problem while they differ in two aspects: the modelling of uncertainty and the mechanism to handle uncertainty. This review aims to draw a methodological picture of historical developments and state-of-the-art advances in this research field. These methods are subsumed into three major categories: multistage optimisation, online optimisation, and multi-timescale optimisation. In addition to introducing the advancements and applications, the authors also explain the motivations and theoretical foundations of these methodologies. The authors hope interested readers can attain a holistic outlook of these sophisticated approaches and choose the most appropriate one for their specific applications.

1 | INTRODUCTION

Energy crisis, environmental pollution, and climate change are three critical issues that the human community is confronting with recently and will have to deal with for the foreseeable future. Major countries in the world have been involved in power grid transformation by integrating more renewable energy generation (REG) to achieve a clean, low-carbon, and sustainable power sector [1], which is acknowledged as a key solution to the issues mentioned above. Despite the global COVID-19 pandemic in 2020, wind power and solar photovoltaics, the two most representative renewable resources, still tend to expand expeditiously with an annual installed capacity growth of 111 and 127 GW, respectively [2].

REG has limited predictability and controllability due to the time-varying weather conditions and thus challenges power system operation from various aspects, such as load balancing, congestion management, stability, and power quality [3]. To support the accommodation of such resources, energy storage units are expected to provide a flexible backup because they can shift demand over time and compensate the real-time mismatch via adjusting charging power and discharging power. Energy storages with various capacities are playing important roles at the generation side, grid side and demand side [4]. According to the analyses and predictions of BloombergNEF, the global energy storage capacity will reach 1676 GW by 2050 [5].

Energy storage dispatch and control with renewable integration cover multiple time slots. At each slot $t \in \mathbb{T}$, the

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decision variables of energy storage include the state of charge (SoC) level E_t and the discharging/charging power P_t^d/P_t^c . For brevity, we use z_t to denote other decision variables in the system, such as nodal voltage and line power flow. The uncertainty of REG and other randomness under consideration are encapsulated in the uncertain parameter ξ_t . Conceptually the system optimal dispatch strategy solves the following problem:

$$\begin{aligned}
 \min \quad & \sum_{t=1}^T f_t(P_t^d, P_t^c, z_t, \xi_t) \\
 \text{s.t.} \quad & E_t = E_{t-1} + \tau \left(P_t^c \eta_c - \frac{P_t^d}{\eta_d} \right), \forall t \in \mathbb{T} \\
 & \underline{P}^c \leq P_t^c \leq \bar{P}^c, \underline{P}^d \leq P_t^d \leq \bar{P}^d, \forall t \in \mathbb{T} \\
 & \underline{E} \leq E_t \leq \bar{E}, \forall t \in \mathbb{T} \\
 & g_t(P_t^d, P_t^c, z_t, \xi_t) \leq 0, \forall t \in \mathbb{T}
 \end{aligned} \tag{1}$$

where τ is the duration of each time period; $\underline{P}^c/\bar{P}^c$ ($\underline{P}^d/\bar{P}^d$) is the lower/upper bound of charging (discharging) power; η_c/η_d is the charging/discharging efficiency; $\underline{E}/\bar{E}$ is the lower/upper bound of the SoC level. The objective function f_t typically reflects system operation cost. Degradation cost of energy storage can also be considered; however, degradation mainly depends on the cycles of charging as well as the depth of discharge of each cycle; such a cost may not have an analytical expression in the decision variables; a rainflow counting-based degradation cost formulation is proposed in Ref. [6]. System operation requirements are imposed in the last constraint. Problem (1) employs a generic model for energy storage [7], which is suitable for battery, pumped hydro storage, and thermal energy storage and alike. However, other storage technologies such as compressed air energy storage may require more sophisticated models. Energy storage modelling is a very important subject, though it is not the main scope of this paper. g_t is a constraint function with regard to network power flow, transmission line capacity etc.

In particular, model (1) does not use binary variables as usual to impose the complementarity condition for charging and discharging states. In most cases, such a condition is naturally satisfied, even it is relaxed. The reason is that it is not an economic strategy due to the energy conversion loss, which is indicated by the charging and discharging efficiency that are less than 100% [8]. Given this fact, the relaxed storage model is widely used, as in Refs. [9, 10]. Meanwhile, to the authors' experience, in some very special problems, binary variables are necessary. In Ref. [11], energy storage is applied to minimise the renewable curtailment; results show that if binary variables are dropped, simultaneously charging and discharging happens, since the energy loss due to efficiency can act as a load that consumes excessive renewable energy and thus reduce the curtailment. Nevertheless, model (1) is applicable in most energy storage dispatch and control problems.

If the parameter sequence $\{\xi_t\}_{t=1}^T$ over the entire operating horizon is known exactly, problem (1) is deterministic and can be solved by commercial solvers. In practice, the following two observations might cause some difficulty in the direct application of such open-loop implementations.

1.1 | Non-anticipativity of decisions

To formulate and solve problem (1), one must have the whole sequence $\{\xi_t\}_{t=1}^T$. The optimal solution can be viewed as a function of $\{\xi_t\}_{t=1}^T$. This means the dispatch strategy in any period t is influenced by the parameters in future periods. However, the realisations of ξ_t are observed period by period in practice; only the value of ξ_t up to the current period is available when the strategy in period t is to be determined. This fact naturally requires that the operation strategy in period t should only depend on the observed uncertainty realisations at present and before, that is, $\{\xi_j\}_{j=1}^t$. Such a requirement is referred to as non-anticipativity, which ensures the causality of decisions [12]. As a result, the decision-making process should be a sequential process.

1.2 | Temporal correlation constraints

If all the constraints of problem (1) are stagewise independent, the problem in each period can be solved separately. However, the charging dynamic equation incorporates both E_{t-1} and E_t , creating the temporal correlation in the storage dispatch and control problems. The temporal correlation introduced by the storage charging dynamics implies that the current decisions will impact the feasible set and the optimal strategy in future periods. Such a feature is indeed a challenge for sequential decision-making because the future is unknown. Generator ramping constraint may be another source of temporal correlation; however, ramping rates of modern gas-fired units are sufficiently large so that the ramping constraints are often redundant.

From the mathematical point of view, energy storage dispatch and control give rise to a sequential decision-making problem with uncertainty and time-couplings, as shown in Figure 1 where x_t is system state such as SoC, y_t is non-state variables such as charging power, and ξ_t is uncertainty. Besides, in problem (1), all the parameters and decision variables are defined in one single time resolution; however, the practice may be much more complex. For example, the wind power profile mainly rests in the frequency band between several minutes and days [13], while the water energy exhibits a much slower seasonal variation influenced by the amount of precipitation. To manage REG variations in the time domain, power system operations include day-ahead and real-time decisions. These strategies are deployed with different timescales. The actions taken in a coarse timescale may influence the finer timescales and vice versa. Therefore, different timescales should be coordinated [14].

Indeed, decision-making under uncertainty is a classical problem that has been studied for a long time. Representative methods include stochastic programming [15, 16] and robust optimisation [17–19]. A comprehensive tutorial on different robust optimisation approaches can be found in Refs. [20, 21]. A recent survey article [22] introduces the advances in multi-stage stochastic and robust optimisation techniques. However, to the best of our knowledge, no review paper specifically focusses on energy storage dispatch and control, which are two very popular topics in power systems with REG integration. Therefore, this paper aims to review and analyse the latest advances regarding this problem. In addition, we offer a tutorial on this timely subject, which we hope can help researchers quickly develop a basic understanding of the state-of-the-art methodologies.

As in Figure 2, this paper reviews the optimisation methods for dispatch and control of energy storage with renewable integration, which mathematically is a sequential decision-making problem involving uncertain parameters and inter-temporal constraints. For single-timescale cases, multi-stage optimisation and online optimisation are discussed; for

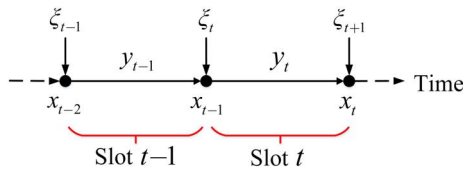


FIGURE 1 Sequential decision-making process

those applications with multiple timescales, multi-timescale optimisation is introduced. The rest of this review paper is organised as follows: in Section 2, multistage optimisation methods are introduced, including the stochastic and robust dynamic programming (DP) as well as a specific discussion on multistage robust unit commitment with energy storage. In Section 3, online optimisation methods are introduced, including Lyapunov optimisation, online convex optimisation, and receding-horizon control. Section 4 reviews multi-timescale optimisation methods. Finally, future directions are pointed out in Section 5.

2 | MULTISTAGE OPTIMISATION

From the mathematical programming point of view, the sequential decision-making process of energy storage dispatch and control can be described by a multistage optimisation problem. The deterministic formulation can be written in a compact form as

$$\begin{aligned} \min \quad & \sum_{t=1}^T c_t^\top x_t + d_t^\top y_t \\ \text{s.t.} \quad & A_t x_t + B_t y_t \leq b_t + F_t x_{t-1}, t = 1 : T \end{aligned} \quad (2)$$

where $x_t \in \mathbb{R}^n, t = 1 : T$ denotes the state variables, and the initial value x_0 is a known constant; $y_t \in \mathbb{R}^m, t = 1 : T$ are non-state variables. Matrices A_t, B_t , and F_t and vectors c_t, d_t , and b_t have compatible dimensions for multiplication. For energy

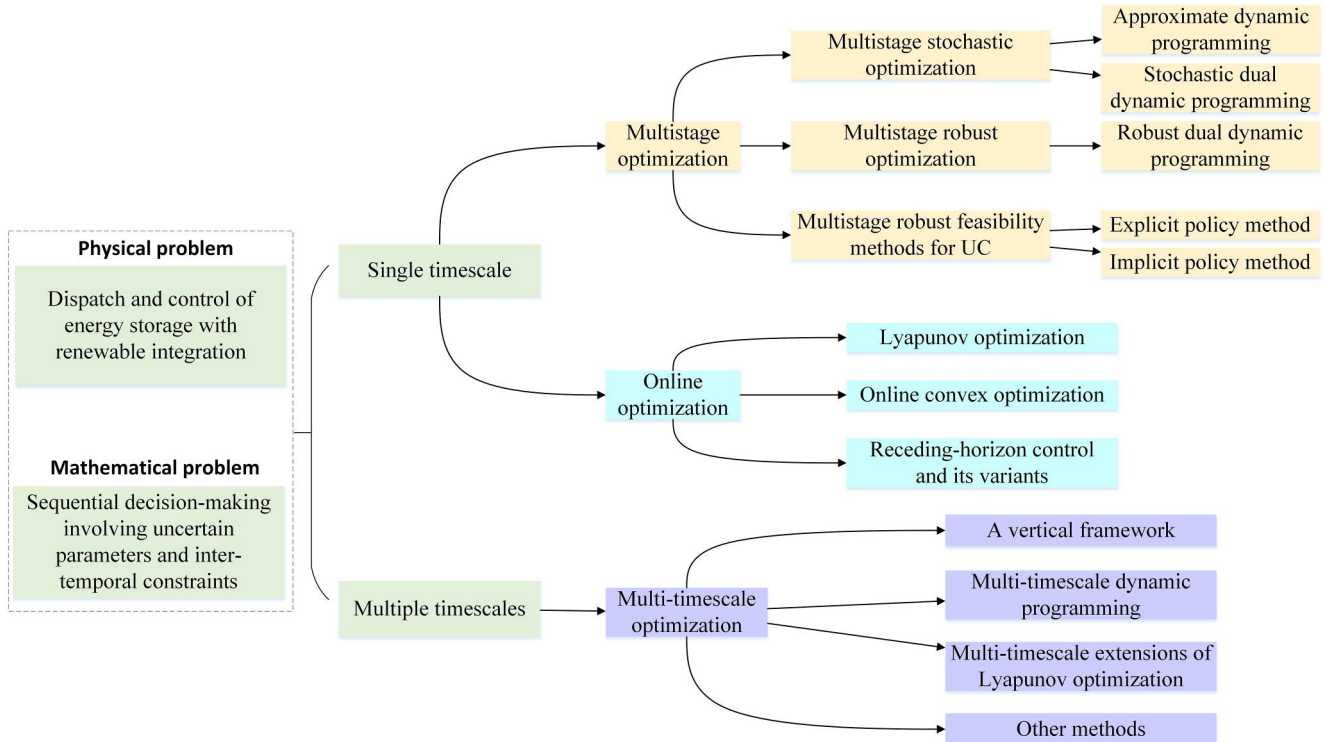


FIGURE 2 Outline of this review

storage, the state transition function $x_t = s(x_{t-1}, y_t)$ is an equality, such as the charging dynamic equation in (1), which can be replaced with two opposite inequalities.

In problem (2), the objective function is stage-wise independent, where the feasible set of (x_t, y_t) depends on x_{t-1} . Consequently, we define

$$\mathcal{X}_t = \{(x_{t-1}, x_t, y_t) | A_t x_t + B_t y_t - F_t x_{t-1} \leq b_t\} \quad (3)$$

and

$$\mathcal{X}_t(x_{t-1}) = \{(x_t, y_t) | A_t x_t + B_t y_t \leq b_t + F_t x_{t-1}\} \quad (4)$$

and rewrite problem (2) in a nested form:

$$\min_{(x_1, y_1) \in \mathcal{X}_1} \left[c_1^\top x_1 + d_1^\top y_1 + \min_{(x_2, y_2) \in \mathcal{X}_2(x_1)} \left[c_2^\top x_2 + d_2^\top y_2 + \dots + \min_{(x_T, y_T) \in \mathcal{X}_T(x_{T-1})} c_T^\top x_T + d_T^\top y_T \right] \right] \quad (5)$$

Using the renowned Bellman's optimality principle in DP [23], problem (5) can be cast recursively by introducing value functions:

$$V_t(x_{t-1}) = \min_{(x_t, y_t) \in \mathcal{X}_t(x_{t-1})} c_t^\top x_t + d_t^\top y_t + V_{t+1}(x_t) \quad (6)$$

where $V_t(x_{t-1})$ is the value function at stage t , representing the future costs related to x_{t-1} . Also, boundary condition $V_{T+1}(x_T) \equiv 0$ is used by default.

Provided with value functions $V_t(x_{t-1})$ and $t = 2 : T$, the optimal solution can be determined sequentially from problem (6), which is decomposed by stage, and the corresponding size is much smaller than that of problem (2). The value functions can be constructed backwards from the last stage; to do this in offline stage, the traditional DP method exhaustively samples x_t and calculates the function value at these points, which are stored for decision-making use.

In DP problem (6), the objective functions are linear and constraints are polyhedral feasible sets, so the value functions are convex and piecewise affine. To see this, we consider that in the last stage $V_{T+1}(x_T) \equiv 0$ and we have

$$V_T(x_{T-1}) = \min_{x_T, y_T} c_T^\top x_T + d_T^\top y_T \quad (7)$$

s.t. $A_T x_T + B_T y_T \leq b_T + F_T x_{T-1} : \lambda_T$

where λ_T is the column vector of dual variables. x_{T-1} is regarded as a parameter and the parameter set is

$$\Theta_{T-1} = \left\{ x_{T-1} \mid \exists (x_T, y_T) : A_T x_T + B_T y_T \leq b_T + F_T x_{T-1} \right\}. \quad (8)$$

The set Θ_{T-1} consists of all the parameter values of x_{T-1} that can achieve a feasible stage T . Clearly, Θ_{T-1} is a polyhedral projection of set $\mathcal{X}_T$ onto the subspace spanned by x_{T-1} , and

thus it is a polyhedron. Polyhedral projection methods can be found in Ref. [24].

For any fixed $x_{T-1} \in \Theta_{T-1}$, problem (7) can be solved, obtaining a unique optimal value, so the optimal value is a function in x_{T-1} . The dual problem of linear programme (7) is

$$V_T(x_{T-1}) = \max_{\lambda_T \in \Lambda_T} \lambda_T^\top (b_T + F_T x_{T-1}) \quad (9)$$

where the dual feasible set is

$$\Lambda_T = \{\lambda_T | A_T^\top \lambda_T = c_T, B_T^\top \lambda_T = d_T, \lambda_T \leq 0\}$$

Strong duality holds for linear programs, so problems (7) and (9) share the same optimal value function. Given any x_{T-1} , the optimal solution of linear programme (9) can always be found at one of the extreme points of set Λ_T , and thus we have

$$V_T(x_{T-1}) = \max_{\lambda_T \in \text{vert}(\Lambda_T)} \lambda_T^\top (b_T + F_T x_{T-1}) \quad (10)$$

where $\text{vert}(\Lambda_T)$ is the vertex set of Λ_T . From (10), the value function $V_T(x_{T-1})$ can be regarded as the point-wise maximum of a finite number of affine functions and thus is convex and piecewise affine. Interested readers are referred to Ref. [25] for more details about the value function; here, we only highlight the convexity and piecewise linearity of value function $V_T(x_{T-1})$ for further induction.

Then, stepping back to stage $T - 1$, by introducing an auxiliary variable δ_T to build the epigraph of $V_T(x_{T-1})$, we have

$$\begin{aligned} V_{T-1}(x_{T-2}) = & \min_{x_{T-1}, y_{T-1}, \delta_T} c_{T-1}^\top x_{T-1} + d_{T-1}^\top y_{T-1} + \delta_T \\ \text{s.t. } & A_{T-1} x_{T-1} + B_{T-1} y_{T-1} \leq b_{T-1} + F_{T-1} x_{T-2} \\ & m_{T-1}^i + n_{T-1}^i x_{T-1} \leq \delta_T, \forall i = 1 : I_{T-1} \\ & x_{T-1} \in \Theta_{T-1} \end{aligned} \quad (11)$$

Problem (11) has the same linear structure as problem (7), and the parameter x_{T-2} appears in the right-hand side coefficient of constraints. Therefore, $V_{T-1}(x_{T-2})$ is also convex and piecewise linear and so do $V_t(x_{t-1})$, $\forall t = 1 : T - 2$ through a backward induction.

Problem (2) is a deterministic linear programme. Given the capability of state-of-the-art solvers, the recursive equation (6) seems unnecessary, especially when the effort in computing the value functions is taken into account. However, when uncertainty is considered, a holistic model such as (2) can hardly be formulated, and the recursive form (6) and value functions are the foundations of multistage decision-making under uncertain conditions. Next, we will discuss several mainstream methods, which differ in modelling of uncertainty, the response to uncertainty, and the derivation techniques for value function.

2.1 | Multistage stochastic programming

Following the paradigm of stochastic programming for modelling the randomness by probability distribution, the multistage stochastic programming problem is formulated as

$$\begin{aligned} \min_{(x_1, y_1) \in \mathcal{X}_1(x_0, \xi_1)} & [f_1(x_1, y_1, \xi_1) + \\ & \mathbb{E}_{\xi_2 | \xi_{[1]}} \left[\min_{(x_2, y_2) \in \mathcal{X}_2(x_1, \xi_2)} [f_2(x_2, y_2, \xi_2) \right. \\ & \left. + \dots + \mathbb{E}_{\xi_T | \xi_{[T-1]}} \left[\min_{(x_T, y_T) \in \mathcal{X}_T(x_{T-1}, \xi_T)} f_T(x_T, y_T, \xi_T) \right] \right] \end{aligned} \quad (12)$$

where $\xi_t, t = 1 : T$ stands for random variables; $\mathcal{X}_t(x_{t-1}, \xi_t)$ is the convex feasible region of (x_t, y_t) , and the cost function f_t is also convex in (x_t, y_t) . The stage-wise dependence of random variables is reflected by the conditional expectations where $\xi_{[t]}$ includes all the realisations up to stage t , that is, $\xi_{[t]} = [\xi_1; \xi_2; \dots; \xi_t]$.

The multistage stochastic programme (12), indicates that at each stage t , after observing the uncertainty realisation $\xi_{[t]}$, the decision (x_t^*, y_t^*) is selected from the set $\mathcal{X}_t(x_{t-1}, \xi_t)$ to minimise the sum of instantaneous cost $f_t(x_t, y_t, \xi_t)$ and the expected future cost. Problem (12) assumes that only consecutive stages can be linked by constraints, which is reasonable in power system operations since both ramping limits and SoC dynamics connect adjacent stages. Besides, problem (12) is assumed to have relatively complete recourse, that is, for any x_{t-1} , the stage t is feasible for any realisation of ξ_t almost surely. This assumption can be ensured by adding slack variables, representing emergency measures such as load shedding and penalising constraint violation in the objective function.

Using the stochastic extension of Bellman's optimality principle, the multistage stochastic programme (12) can be reformulated in a recursive manner into a stochastic DP problem, that is,

$$V_t(x_{t-1}, \xi_t) = \min_{(x_t, y_t) \in \mathcal{X}_t(x_{t-1}, \xi_t)} f_t(x_t, y_t, \xi_t) + \mathbb{E}_{\xi_{t+1} | \xi_{[t]}} V_{t+1}(x_t, \xi_{t+1}) \quad (13)$$

where $V_t(x_{t-1}, \xi_t)$ is the value function (or cost-to-go function) at stage t parameterised in x_{t-1} and ξ_t with boundary condition $V_{T+1}(x_T, \xi_{T+1}) \equiv 0$ by default. If functions $V_{t+1}(x_t, \xi_{t+1})$ and $t = 1 : T$ and the conditional probability distribution of ξ_{t+1} are known, the decisions at each stage can be obtained by solving problem (13).

However, computing all the exact value functions suffers from the 'curses of dimensionality' [26]. To compute $V_t(x_{t-1}, \xi_t)$, we have to exhaustively visit every possible combination of (x_{t-1}, ξ_t) in and loop over problem (13); further, the stage-wise dependence makes it more difficult to calculate the expectation in (13). A natural way to get released from these curses is to approximate the value functions. Next, we introduce two typical methods.

2.1.1 | Approximate dynamic programming

The most straightforward method is approximate dynamic programming (ADP). There are two main types of ADP algorithms: backward-type and forward-type, which means they approximate value functions by backward and forward recursions, respectively. The backward-type ADP is often cursed by dimensionality, although it may work in some applications such as residential energy management, where the state space has a dimension of only one or two [27]. In addition to the above two prevalent methods, some ADP techniques aim to approximate the policy functions, which are explicit decision rules. These rules directly offer an action that relies on up-to-date uncertainty realisations and system state. Table 1 collects a group of research papers on ADP and its applications in energy system, especially those which are on the operation of storage-integrated systems.

In the ADP community, the forward-type value function approximation (VFA) is the most widely used tool, which is comprehensively introduced by W. Powell in his famous book [26]. A general framework of forward ADP is given in Algorithm 1, which is developed in figure 4.2 in Ref. [26]. This general algorithm is not executable, but it reveals the core idea of forward ADP: it steps forwards from stage 1 to stage T iteratively; in iteration n , problem (13) is built with the value function obtained in iteration $n - 1$. In step (2b), the value functions are updated. Accordingly, several approaches are introduced in Ref. [26], such as look-up table, regression, learning, and so on to perform this function as part of ADP. In particular, the learning technique and ADP are closely related; both of them learn and adapt in the offline training phase for online execution [32].

Algorithm 1 Approximate dynamic programming framework

Initiation: Initialise function $V_{t+1}^0(x_t, \xi_{t+1})$ for $t = 1 : T$; Sample N path $\{\xi_t^n\}_{t=1}^T, n = 1 : N$; choose an initial state x_1^0 and set the iteration index $n = 1$.

Step 1. Choose a path $\xi^n = \{\xi_1^n, \dots, \xi_T^n\}$.

Step 2. For $t = 1 : T$

(2a). Solve problem (13) with x_{t-1}^n, ξ_t^n , and $V_{t+1}^{n-1}(x_t, \xi_{t+1})$; save the optimal value v_t^n and optimal solution x_t^n .

(2b). Update $V_t^{n-1}(x_{t-1}, \xi_t)$ to $V_t^n(x_{t-1}, \xi_t)$ with (x_{t-1}^n, v_t^n) .

End

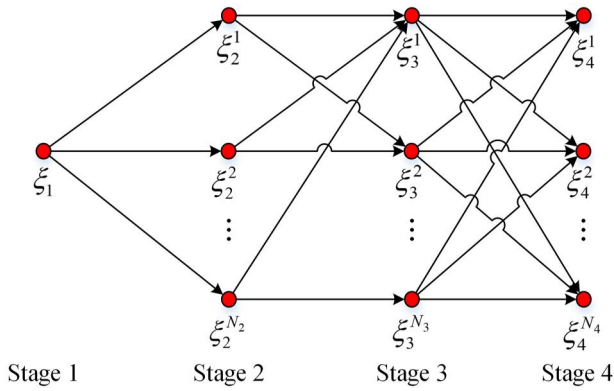
Step 3. Let $n \leftarrow n + 1$. If $n < N$, go back to step 1.

Output: Approximate value functions $V_{t+1}(x_t, \xi_t), t = 1 : T$.

TABLE 1 Research papers of approximate dynamic programming used in energy system applications

Reference	Application	Main technical innovation
[27]	Residential battery storage operations	Distributionally robust data-driven VFA [28]
[29]	Wind-storage system operation	Linear interpolation for VFA
[30]	Residential prosumer energy management	Multiparametric programming for VFA with lower bound guarantee
[31]	Microgrid economic dispatch	A modified slope updating method for VFA that speeds up the convergence
[32]	Microgrid energy management	Deep recurrent neural network learning for VFA
[33]	Gas-power network constrained unit commitment	Slope update with difference and derivative for ADP with binary decisions
[34]	Photovoltaic-storage system energy management	Machine learning to train policy function
[35]	Energy hub dispatch	Parametric cost function approximation

Abbreviations: ADP, approximate dynamic programming; VFA, value function approximation.

**FIGURE 3** Scenario tree

2.1.2 | Stochastic dual dynamic programming

It's generally acknowledged that Ref. [26] is an excellent textbook for forward ADP researchers and users. In view of this, we pay more attention to another popular approach that iteratively approximates the value functions, called stochastic dual dynamic programming (SDDP). SDDP is substantially a member of ADP community since it also creates approximate value functions for stochastic dynamic programming problem (23). However, SDDP is different from ADP in many respects, including how they sample and use data and how they approximate value functions [36].

Traditional SDDP assumes the stage-wise independence of random variables and considers a stochastic linear DP problem

$$\begin{aligned}
 V_t(x_{t-1}, \xi_t) \\
 = \min_{x_t, y_t} \quad & c_t^\top(\xi_t)x_t + d_t^\top(\xi_t)y_t + \mathbb{E}_{\xi_{t+1}}[V_{t+1}(x_t, \xi_{t+1})] \\
 \text{s.t.} \quad & A_t(\xi_t)x_t + B_t(\xi_t)y_t \leq b_t(\xi_t) + F_t(\xi_t)x_{t-1} : \lambda_t(\xi)
 \end{aligned} \quad (14)$$

where all the coefficients are likely to be affected by random variable ξ_t . With $V_{T+1}(x_T, \xi_{T+1}) \equiv 0$, any $V_t(x_{t-1}, \xi_t)$, $t = 1 : T$ for a fixed ξ_t is piecewise affine and convex in x_{t-1} , which can be easily proven by the same arguments as those in (7)–(11).

The information structure of problem (14) is illustrated via a scenario tree. If uncertainty ξ_t is discrete, the scenario tree in stage t has N_t possible outcomes; if ξ_t is a continuous variable, sample average approximation is employed to generate a finite tree [37]. Under the stage-wise independence assumption, a 4-stage scenario tree is shown in Figure 3. The root node is deterministic, so we do not need $V_1(x_0, \xi_1)$. Consequently, problem (14) can be reformulated as

$$\begin{aligned}
 V_t(x_{t-1}, \xi_t) \\
 = \min_{x_t, y_t} \quad & c_t^\top(\xi_t)x_t + d_t^\top(\xi_t)y_t + \frac{1}{N_t} \sum_{n=1}^{N_t} [V_{t+1}(x_t, \xi_{t+1}^n)] \\
 \text{s.t.} \quad & A_t(\xi_t)x_t + B_t(\xi_t)y_t \leq b_t(\xi_t) + F_t(\xi_t)x_{t-1} : \lambda_t(\xi)
 \end{aligned} \quad (15)$$

where the expected future costs are approximated by the average of sampled scenarios. For brevity, we use the set Ξ_t to represent the nodes at stage t in the scenario tree, that is, Ξ_t . The rest is to calculate $V_{t+1}(x_t, \xi_{t+1})$, $\forall \xi_{t+1} \in \Xi_{t+1}$, $t = 1 : T$.

Before presenting the conventional SDDP algorithm, we explain what this algorithm would do, which might, to a great extent, help understand how it works. Given any $\bar{\xi}_t \in \Xi_t$, $V_t(x_{t-1}, \bar{\xi}_t)$ is convex and piecewise affine in x_{t-1} as claimed above. If it is able to visit every feasible x_{t-1} , we can build the function $V_t(x_{t-1}, \bar{\xi}_t)$ with every pair of x_{t-1} and its resulting optimal value of problem (15). However, in general cases, x_{t-1} is continuous and visiting an infinite number of points is not practical. So, we can visit only a few of them to build the complete $V_t(x_{t-1}, \bar{\xi}_t)$ with limited information at some points of x_{t-1} . The flowchart of SDDP, which performs such function is summarised in Algorithm 2. The forward pass strategically selects the points to be visited, and the backward pass updates the value functions via Benders cuts at the visited points.

Algorithm 2 Stochastic dual dynamic programming [38]

Initiation: Function $V_{T+1}(x_T, \xi_{T+1}) \equiv 0$; a fixed initial state x_0 ; a scenario tree with

T stage, whose t -stage node set is Ξ_t . Set the t -stage cut index $I_t = 0$;

Sampling: Choose from the scenario tree N paths, that is, $\{\xi_t^n\}, t = 1 : T, n = 1 : N$.

Forward pass:

For $t = 1 : T$,
For $n = 1 : N$
 Solve problem

$$\begin{aligned} \min_{x_t, y_t, \theta_{t+1}} \quad & c_t^\top(\xi_t^n)x_t + d_t^\top(\xi_t^n)y_t + \theta_{t+1} \\ \text{s.t.} \quad & A_t(\xi_t^n)x_t + B_t(\xi_t^n)y_t \leq b_t(\xi_t^n) + F_t(\xi_t^n)x_{t-1}^{n*} \\ & \theta_{t+1} \geq k_{t+1}^i x_t + b_{t+1}^i, i \leq I_t \\ & \theta_{t+1} \geq 0 \end{aligned} \quad (16)$$

Save optimal solution x_t^{n*} .

End

End

Convergence criterion: Please see remark.

Backward pass:

For $t = T : 2$
For $n = 1 : N$
For $\xi_t^s \in \Xi_t$
 Solve problem

$$\begin{aligned} \min_{x_t, y_t, \theta_{t+1}} \quad & c_t^\top(\xi_t^s)x_t + d_t^\top(\xi_t^s)y_t + \theta_{t+1} \\ \text{s.t.} \quad & A_t(\xi_t^s)x_t + B_t(\xi_t^s)y_t \leq b_t(\xi_t^s) + F_t(\xi_t^s)x_{t-1}^{n*} : \lambda_t(\xi_t^s) \\ & \theta_{t+1} \geq k_{t+1}^i x_t + b_{t+1}^i, i \leq I_t \\ & \theta_{t+1} \geq 0 \end{aligned} \quad (17)$$

Save dual solution λ_t^{ns*} and optimal value v_t^{ns*} ;

End

Update $I_{t-1} \leftarrow I_{t-1} + 1$

Build the I_{t-1} -th cut for x_{t-1} :

$$\theta_t \geq k_t x_{t-1} + b_t \quad (18)$$

where

$$k_t = \frac{\sum_{s=1}^{|\Xi_t|} [(\lambda_t^{ns*})^\top F_t(\xi_t^s)]}{|\Xi_t|}, b_t = \frac{\sum_{s=1}^{|\Xi_t|} v_t^{ns*}}{|\Xi_t|} - k_t x_{t-1}^{n*}$$

End

End

Go back to forward pass.

Output: Approximate value functions

$V_{t+1}(x_t, \xi_t^n), t = 1 : T, n = 1 : N$.

Here is a remark for the stopping criterion of SDDP. Since we use Benders cuts to approximate the value function, we have a valid lower bound $\underline{v}$ in each iteration that is obtained by solving problem (16) in stage 1. Meanwhile, we have a statistical upper bound estimated by $\bar{v} = \frac{1}{N} \sum_{n=1}^N \bar{v}^n$ where $\bar{v}^n = \sum_{t=1}^T c_t^\top(\xi_t^n)x_t^{n*} + d_t^\top(\xi_t^n)y_t^{n*}$. A confidence criterion is developed in Ref. [39], which identifies whether the lower bound $\underline{v}$ falls into a confidence interval $[LB, RB]$, which is built by $(x_t^{n*}, y_t^{n*}), t = 1 : T, n = 1 : N$. This criterion may favour premature stopping, since the true upper bound is unlikely to be located exactly at LB . So, alternative criteria are put forward. For those problems that are too large to identify convergence within a reasonable time, it may be convenient to stop after a certain number of iterations. Then, a more conservative criterion is devised in Ref. [37]; if the difference between $\underline{v}$ and RB is small enough, the SDDP algorithm will be terminated. Besides, a modified hypothesis test is proposed in Ref. [40] to compromise conservatism and optimism about judging convergence. Nevertheless, the criterion for terminating SDDP algorithms remains an open problem.

SDDP is first developed in Ref. [39] where this algorithm is applied to optimally schedule hydro reservoirs; since that, many studies discuss SDDP at the algorithmic level. In Ref. [37], the SDDP solutions with sample average approximation and their relations to the true problem are analysed, and an extension of SDDP is developed to formulate a risk-averse multistage stochastic programme. The risk-averse version of SDDP is also investigated in Ref. [41]. It is found in Ref. [38] that increasing the number of scenarios in forward pass may significantly improve the performance of SDDP. The convergence of SDDP and the related methods is proven in Ref. [42]. The complexity of SDDP is discussed in Ref. [43]; the results indicate that the complexity mildly increases with the number of stages, so SDDP is efficient for a problem that involves a moderate number of decision variables at each stage but includes a large number of stages. As a remedy of using the SAA formulation, the distributionally robust SDDP is proposed Ref. [44] to consider the inexactness of uncertainty probability distribution with a distributional uncertainty set, which is a Euclidean neighbourhood of the empirical probability distribution. To handle the non-continuity in multistage stochastic programs with integer decision variables, X. Sun et al. developed stochastic dual dynamic integer programming in Ref. [45] and its extension to non-linear problems in Ref. [46]. A variant of SDDP is proposed by Sen in Refs. [47, 48], which uses sequential sampling technique and thus does not necessitate a scenario tree that requires a priori knowledge of the probability distribution. Besides, a review of SDDP and its variants is found in Ref. [49].

The SDDP framework has been applied in power systems and energy storage optimisation problems with REGs. In large power systems, the real-time economic dispatch with pumped hydro storages is formulated in Ref. [50] as a multistage

stochastic programme and solved by SDDP. The joint energy and reserve dispatch via SDDP is reported in Ref. [51]; specifically, the multistage stochastic programming solved by SDDP is compared with two-stage stochastic programming and the results show that SDDP achieves a significantly lower cost while it consumes more computation time, since it entails several rounds of forward–backward recursions. In small-scale cases, the energy storage management problem in a grid-connected microgrid is studied in Ref. [52] using a customised SDDP; a dynamic cut selection procedure and a lower bound improvement scheme refine the performance of standard algorithm. The operation of distributed energy resource aggregators under multi-dimensional uncertainty is scheduled by SDDP in Ref. [53]; to capture the dependencies among different stages and uncertain variables, a vector autoregressive model is integrated into the SDDP algorithm, which depicts the evolution of a multivariate stochastic process as an affine function of previous realisations. Besides, the aforementioned stochastic dual dynamic integer programming is employed in Ref. [54] for multistage stochastic unit commitment. One difficulty is that the stochastic dual dynamic integer programming requires all the state variables to be binary; as a remedy, the continuous generation variables are discretised by the binary approximation technique. Eventually, this method offers a multistage policy that operators can use in the real-time commitment and dispatch.

To summarise, the SDDP framework is very effective in energy storage dispatch and control and power system operation, which releases the curses of dimensionality by strategic value function approximation. This is very important because in large power systems; both the power production of thermal units and the SoC level of energy storages are state variables, so the dimension of state space is very high. Meanwhile, the uncertainty is described by the scenario tree, which can be handily constructed by historical data. However, this method relies on the stage-wise independence assumption, which is sometimes oversimplified; the dependence between stages may include some information of uncertainty evolution that can improve the performance of stochastic DP. Besides, the approximated value functions may not be robust against some extreme scenarios of uncertainty realisation, which is a common weakness of the methods in the stochastic programming community.

2.2 | Multistage robust optimisation

Evaluating a high-quality probability distribution entails a significant amount of historical data, which may not be available at hand. In such circumstances, the uncertainty of data variation can be alternatively described by a set as that in robust optimisation, which is formally named an uncertainty set. Consequently, at each stage t , the decisions (x_t, y_t) are taken with x_{t-1} after the value of ξ_t is observed, aiming to minimise the sum of instantaneous and future costs in the worst-case scenario. This gives rise to the following multistage robust optimisation problem:

$$\begin{aligned} \min_{(x_1, y_1) \in \mathcal{X}_1(x_0, \xi_1)} & \left[c_1^\top(\xi_1)x_1 + d_1^\top(\xi_1)y_1 \right. \\ & + \max_{\xi_2 \in \Xi_2} \left[\min_{(x_2, y_2) \in \mathcal{X}_2(x_1, \xi_2)} \left[c_2^\top(\xi_2)x_2 + d_2^\top(\xi_2)y_2 \right. \right. \\ & \left. \left. + \cdots + \max_{\xi_T \in \Xi_T} \left[\min_{(x_T, y_T) \in \mathcal{X}_T(x_{T-1}, \xi_T)} c_T^\top(\xi_T)x_T + d_T^\top(\xi_T)y_T \right] \right] \right] \end{aligned} \quad (19)$$

where

$$\Xi_t = \{\xi_t | H_t \xi_t \leq h_t\} \quad (20)$$

is the polyhedral uncertainty set of ξ_t and

$$\mathcal{X}_t(x_{t-1}, \xi_t) = \left\{ (x_t, y_t) \left| \begin{array}{l} A_t(\xi_t)x_t + B_t(\xi_t)y_t \\ \leq b_t(\xi_t) + F_t(\xi_t)x_{t-1} \end{array} \right. \right\}. \quad (21)$$

is the feasible region of (x_t, y_t) relying on x_{t-1} and ξ_t .

Similarly, we build the recursive form of the multistage robust optimisation problem (19) as follows:

$$\begin{aligned} V_t(x_{t-1}, \xi_t) = & \min_{(x_t, y_t) \in \mathcal{X}_t(x_{t-1}, \xi_t)} \left[c_t^\top(\xi_t)x_t + d_t^\top(\xi_t)y_t \right. \\ & \left. + \max_{\xi_{t+1} \in \Xi_{t+1}} V_{t+1}(x_t, \xi_{t+1}) \right] \end{aligned} \quad (22)$$

where $V_t(x_{t-1}, \xi_t)$ is still the value function at stage t with terminal condition $V_{T+1}(x_T, \xi_{T+1}) \equiv 0$. To facilitate solution, we define the worst-case value function

$$Q_{t+1}(x_t) = \max_{\xi_{t+1} \in \Xi_{t+1}} V_{t+1}(x_t, \xi_{t+1})$$

which is a function of x_t . If we have $Q_{t+1}(x_t)$ in advance, the strategy at stage t is the optimiser of

$$\min_{(x_t, y_t) \in \mathcal{X}_t(x_{t-1}, \xi_t)} \left[c_t^\top(\xi_t)x_t + d_t^\top(\xi_t)y_t + Q_{t+1}(x_t) \right] \quad (23)$$

Because both x_t and ξ_{t+1} appear in the right-hand coefficient of constraints, following the same arguments as those in (7)–(11), it can be shown that $V_{t+1}(x_t, \xi_{t+1})$ is convex in x_t and ξ_{t+1} . As a result, $Q_{t+1}(x_t)$ entails a concave programme over a polyhedron Ξ_{t+1} , and the optimal solution must be achieved at some vertex of Ξ_{t+1} [55, 56]. Let $\text{vert}(\Xi_{t+1}) = \{\xi_{t+1}^{i1}, \dots, \xi_{t+1}^{im}\}$, then

$$Q_{t+1}(x_t) = \max \left\{ V_{t+1}^j(x_t, \xi_{t+1}^{ij}) \right\}$$

implying that $Q_{t+1}(x_t)$ is the pointwise maximum of convex piecewise affine functions and thus remains convex and piecewise affine. In principle, if the dimension of ξ_t is low, it is possible to compute the analytical expressions of

$\left\{V_{t+1}^j(x_t, \xi_{t+1}^{ij})\right\}_{j=1}^m$. Nevertheless, a strategy similar to SDDP can be applied, motivating the so-called robust dual dynamic programming (RDDP) algorithm [57]. RDDP has a similar forward-backward recursive scheme where the forward pass also selects the interesting points x_t to be visited, while the backward pass is somewhat different from that in SDDP. Particularly, the backward pass of RDDP generates a lower-bound function $\bar{Q}_{t+1}(x_t)$ and an upper-bound function $\underline{Q}_{t+1}(x_t)$, to squeeze an approximation of $Q_{t+1}(x_t)$. These two functions are, respectively, constructed by Benders cuts and convex hull with respect to a limited number of visited points.

The RDDP algorithm has been applied in some energy storage dispatch and control problems, including the energy management of a storage-based residential prosumer in Ref. [58] and microgrids in Ref. [59]. Compared to SDDP, RDDP reduces the computational burden since it uses the uncertainty set instead of the scenario tree to describe the stochasticity. However, we find that formulation (19) considers the worst-case scenario of uncertainty realisation at each stage, which may be conservative.

2.3 | Multistage robust unit commitment with energy storage: A feasibility perspective

Unit commitment is one of the most fundamental problems of power system operation, which underlies many other daily operation problems such as economic dispatch and optimal power flow. This important problem is worthy of a particular discussion in this review. A unit commitment solution, which usually is determined day-ahead, indicates the on/off state of the thermal units in each hour of the next day, based on which the economic dispatch is sequentially deployed. This is clearly a multistage process.

2.3.1 | Two-stage or multi-stage?

To immunise against the impact of uncertainty on the optimality and feasibility of economic dispatch, the two-stage robust unit commitment formulation in Ref. [60] is widely adopted, which is

$$\begin{aligned} \min_x \quad & \left(c^\top x + \max_{\xi \in \Xi} \min_{y \in \Omega(x, \xi)} b^\top y \right) \\ \text{s.t.} \quad & Fx \leq f, x \text{ is binary} \end{aligned} \quad (24)$$

where x encapsulates the hourly on/off state of thermal units. ξ is the uncertainty vector of renewable power, which is constrained within a polyhedral uncertainty set Ξ . y is the vector of economic dispatch variables, whose feasible region is denoted as $\Omega(x, \xi)$. Energy storage can be modelled in several ways, one of which is to regard it as a flexible unit without ramping limits. The charging/discharging state is included in x and decided

ahead of the day, and the hourly charging/discharging power is included in y [61]. x , ξ , and y are all column vectors and aggregate all the stagewise vectors. Problem (24) is two-stage adjustable robust optimisation problem [20].

The decision-making logic of the two-stage robust unit commitment can be interpreted as follows: the first stage action x^* minimises the sum of startup/shutdown costs and the dispatch costs under the worst-case realisation of uncertainty ξ^* ; the optimal second-stage action y^* is taken after observing x^* and ξ^* in order. As a result, x^* is a two-stage robust unit commitment solution, which is optimal under the worst-case realisation ξ^* and (two-stage) feasible for any other realisation of $\xi \in \Xi$, which means given x^* , we can find at least one feasible solution $y \in \Omega(x^*, \xi)$ for any $\xi \in \Xi$; this is achieved by simultaneously generating optimality and feasibility cuts via Column-and-Constraint Generation (C&CG) algorithm [62].

However, if the multistage property of economic dispatch is taken into account, practical issues arise in (24). y/ξ includes the uncertainty/dispatch variables for a full day; y is determined after ξ is observed, indicating that the dispatch decisions are anticipative. The consequence is that the two-stage robust unit commitment solution x^* , which is feasible in two-stage settings, might be infeasible in the multistage problem [63]. That is, supposing the unit commitment x^* is acquired from (24); if we sequentially schedule $y_t \in \Omega_t(x^*, y_{t-1}, \xi_t)$ for t from 1 to T with uncertainty realisations ξ_t , the process might not be able to proceed in slot t' because $\Omega_{t'}(x^*, y_{t-1}, \xi_{t'}) = \emptyset$ for some $\xi_{t'} \in \Xi_{t'}$.

For further illustration, consider the following example. Suppose a stand-alone energy system consisting of one thermal unit, one energy storage, and one load: in every time slot $t \in T = \{1, 2\}$, the generation of thermal unit is constrained by $1 \leq P_t^G \leq 6, \forall t \in T$; the ramping limits is 1, that is, $\|P_2^G - P_1^G\| \leq 1$. The allowable range of charging/discharging power is $3 \leq P_t^C/P_t^D \leq 6, \forall t \in T$. The demand in the first slot is $D_1 = 7$ while in the second slot is $D_2 = 10 - 6k$ with a random variable $k \in [0, 1]$. For simplicity, all the variables are denoted unitless, and energy storage is lossless.

Regarding the energy storage as a unit, consider the unit commitment solution where the thermal unit is on and energy storage operates in discharging mode for every time slot, denoted by UC^* . Then, we step into the economic dispatch phase with UC^* .

1) In the two-stage robust setting, $\xi = [D_1, D_2]^\top$ is prior to $y = [P_1^G, P_2^G, P_1^D, P_2^D]^\top$. Let $Policy^{ts}$ be

$$Policy^{ts} : \begin{cases} P_1^D = 4 + k \\ P_1^G = 3 - k \\ P_2^D = 6 - 3k \\ P_2^G = 4 - 3k \end{cases} \quad (25)$$

Since $k \in [0, 1]$, it is easy to check that the policy in (25) falls into the allowable range of P_t^G and P_t^D for any $t \in T$; meanwhile, we have $\|P_2^G - P_1^G\| = \|1 - 2k\| \leq 1$. Also,

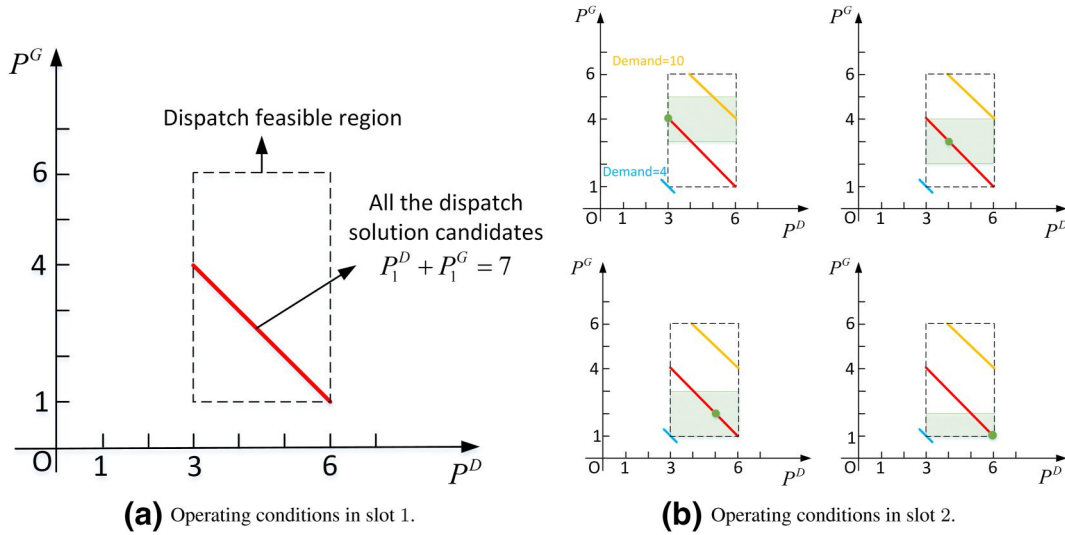


FIGURE 4 Operating condition analysis of the illustrative example in a multistage manner

$P_1^D + P_1^G = D_1$ and $P_2^D + P_2^G = D_2$, so the demand can be satisfied. Therefore, for any $k \in [0, 1]$, we have at least one feasible economic dispatch solution given by (25), verifying that UC^* is a two-stage robust feasible unit commitment solution.

- 2) In multistage settings, the economic dispatch decisions are non-anticipative, so P_1^D and P_1^G cannot rely on k . With UC^* , we revisit the economic dispatch phase in a multi-stage manner.

In slot 1, the power balance condition must be satisfied, so we have $P_1^D + P_1^G = 7$. As graphically shown in Figure 4a, the dotted-line rectangle is the feasible region of P_1^D and P_1^G ; all the dispatch solution candidates in the first slot that can ensure power balance constitute the red line segment.

Given any solution in the first slot (green dot in Figure 4a), all the reachable solutions in the second slot are covered by the shaded rectangle in Figure 4b. Taking the first subfigure as an example, from the slot 1 solution $(P_1^D, P_1^G) = (3, 4)$, the discharging power of energy storage in the second slot can be freely chosen from interval $[3, 6]$; however, the thermal unit generation in the second slot is constrained within $[3, 5]$ due to the ramping limit, giving rise to the shaded rectangle feasible region. At the same time, D_2 is a random variable that varies within $[4, 10]$. From this subfigure, it can be seen that when $D_2 \in [4, 6)$, there is no feasible slot 2 solution that can fulfil power balance. Similar inferences can be drawn from the other three subfigures, and finally, it is not difficult to conclude that no matter what solution in slot 1 is made without knowing D_2 , there must be some D_2 that leads to an infeasible dispatch in the second slot. Therefore, UC^* is NOT a multi-stage robust feasible unit commitment solution.

The above example signifies that a two-stage robust feasible unit commitment solution may be infeasible in a multi-

stage problem, because the two-stage robust feasibility of x^* relies on the availability of uncertainty in all periods at once and thus is substantially optimistic. To address this issue, consider the following multistage robust unit commitment formulation:

$$\begin{aligned} \min_x \quad & c^T x + \max_{\xi \in \Xi} \min_y b^T y \\ \text{s.t.} \quad & Fx \leq f, x \text{ is binary} \\ & y \in \Omega(x, \xi), \forall \xi \in \Xi \\ & y_t = f_t(\xi_{[t]}), t = 1 : T \end{aligned} \quad (26)$$

where $\xi_{[t]}$ represents the uncertainty realisations up to slot t , that is, $\xi_{[t]} = [\xi_1; \dots; \xi_t]$. The implicit function $f(\xi_{[t]})$ stipulates that the dispatch decision y_t can only depend on $\xi_{[t]}$, satisfying the non-anticipativity condition. According to the current research, the non-anticipativity requirement can be implemented via either explicit policy or implicit policy.

2.3.2 | Explicit policy

The idea of explicit policy was first suggested by Ben-Tal et al. in Ref. [64], where the adjustable variables in adjustable robust optimisation are restricted to be affine functions of the uncertainty. Considering the multistage decision-making paradigm with non-anticipativity, we set the economic dispatch decisions y_t to be affine in the uncertainty realisations up to slot t , that is,

$$y_t = K_t \xi_{[t]} + B_t, \forall t \quad (27)$$

where K_t/B_t is a matrix/column vector to be optimised. With a direct-current power flow model, the second constraint in (26) is linear, that is, $\Omega = \{y | My \leq m(x, \xi)\}$. Replacing y_t with its affine equivalence in (27), problem (26) turns into

$$\begin{aligned}
& \min_{x, K, B} \left(c^\top x + \max_{\xi \in \Xi} b^\top \begin{bmatrix} K_1 \xi_{[1]} + B_1 \\ \vdots \\ K_T \xi_{[T]} + B_T \end{bmatrix} \right) \\
& \text{s.t. } Fx \leq f, x \text{ is binary} \\
& M \begin{bmatrix} K_1 \xi_{[1]} + B_1 \\ \vdots \\ K_T \xi_{[T]} + B_T \end{bmatrix} \leq m(x, \xi), \forall \xi \in \Xi
\end{aligned} \quad (28)$$

where $K = [K_1; K_2; \dots; K_T]$ and $B = [B_1; B_2; \dots; B_T]$ are decision variables.

Although problem (28) still has a two-stage structure, it can generate a multistage robust feasible unit commitment solution, because the economic dispatch problem is feasible in any uncertainty realisation $\xi \in \Xi$ under affine policy (27), which guarantees the non-anticipativity of dispatch decisions. In practice, problem (28) is solved prior to the operating day; then, the unit commitment solution x^* is deployed, and the explicit policy coefficients K and B are passed to the dispatcher, which sequentially deploys the policy y_t with $\xi_{[t]}$, the uncertainty realisations that have been collected.

The remaining task is to solve problem (28), which is a so-called semi-infinite programme, that is, there are finite decision variables but an infinite number of constraints. Two effective methods are introduced in Ref. [63]. One is the duality-based approach, which reformulates the robust constraints in (28) into their robust counterpart via linear programming duality. The other, named constraint generation, is designed to identify a finite number of extreme points of set Ξ to build a tractable deterministic programme.

The explicit policy method is also discussed in Ref. [65], which has two differences. One is that the expected dispatch costs of several typical uncertainty scenarios, instead of the worst-case scenario costs, are minimised in the multistage robust unit commitment formulation, which reduces the conservatism in optimality. The other is that the robust constraints are substituted with their equivalent or sufficient conditions given a box uncertainty set.

A simplified affine policy is defined in Ref. [66], saying that the dispatch decision y_t is affine only to ξ_t instead of $\xi_{[t]}$, that is, $y_t = K_t \xi_t + B_t, \forall t$; nevertheless, the solution methods still work. The significance of explicit policy under the adjustable robust framework is not only defining the non-anticipative decisions to achieve multistage robust feasibility but also mapping the recourse actions from uncertain data in an analytical manner. Analytical functions are appealing for real-time applications since they are able to reduce the computational burden and thus release the requirement for hardware. Real-time dispatch and optimal power flow with affine policy are studied in Refs. [67, 68]. The suboptimality and the loss of adaptability due to the restriction of redispatch policy to be affine were discussed in Refs. [69, 70].

In addition to the affine policy, other explicit decision rules have been reported in the literature, such as quadratic rule [17], piecewise constant rule [71], piecewise affine rule [72] etc. These rules lead to more complicated optimisation problems;

for example, semi-definite programs are for quadratic rules, and second-order cone programs are for piecewise affine rules. The performance of these rules often depends on specific scenarios.

2.3.3 | Implicit policy

Although the explicit policy method shows some advantages in effectiveness and tractability, it more or less sacrifices some adaptability in economic dispatch stage. Instead of restricting the functions between dispatch actions and uncertain data, the implicit policy method explores another avenue by defining a specific feasible region for dispatch variables y_t in each slot t . Let $\Omega_t(x^*, y_{t-1}, \xi_t)$ be the original feasible set of y_t ; if we want Ω_t to be non-empty given any ξ_t, x^* and y_{t-1} must be carefully chosen. To this end, two major steps are taken in formulating the multistage robust unit commitment with implicit policy:

- (1) Selecting vertex scenarios [73]: Consider a box uncertainty set Ξ describing the power of K renewable plants in T dispatch slots. In any slot t , the allowable power range of the k th plant is $[\underline{\xi}_{k,t}, \bar{\xi}_{k,t}]$. Therefore, in slot t we have a polyhedral uncertainty subset Ξ_t that has 2^K vertices. One selected vertex scenario of Ξ aggregates one vertex in each Ξ_t in parallel. For clarification, we assume a system with two renewable plants operating in T time slots, that is, $K = 2$. For each $t \in \{1, \dots, T\}$, we have $2^2 = 4$ vertices of uncertainty subset Ξ_t :

$$\begin{aligned}
\xi_t^1 &= (\underline{\xi}_{1,t}, \underline{\xi}_{2,t}), \xi_t^2 = (\underline{\xi}_{1,t}, \bar{\xi}_{2,t}) \\
\xi_t^3 &= (\bar{\xi}_{1,t}, \underline{\xi}_{2,t}), \xi_t^4 = (\bar{\xi}_{1,t}, \bar{\xi}_{2,t})
\end{aligned} \quad (29)$$

Aggregating one vertex in each Ξ_t in parallel to develop one vertex scenario for Ξ can be expressed as follows:

$$\begin{aligned}
\xi^1 &= [\xi_1^1, \xi_2^1, \dots, \xi_T^1]^\top, \xi^2 = [\xi_1^2, \xi_2^2, \dots, \xi_T^2]^\top, \\
\xi^3 &= [\xi_1^3, \xi_2^3, \dots, \xi_T^3]^\top, \xi^4 = [\xi_1^4, \xi_2^4, \dots, \xi_T^4]^\top
\end{aligned} \quad (30)$$

This parallel vertex scenario selection is depicted in Figure 5. Therefore, the number of selected vertex scenarios of Ξ is also 2^K . The role of selected vertex scenarios is to guarantee the robust feasibility in every single slot; if the system is feasible in slot t to every vertex of Ξ_t , it is feasible to every realisation $\xi_t \in \Xi_t$ because there is at least one feasible solution to ξ_t that can be expressed by the convex combination of the solutions to all vertices. Please refer to Appendix B in Ref. [73] for details.

- (2) Constructing non-anticipativity constraints: Non-anticipativity constraints are designed to handle the time-couplings in economic dispatch where the ramping limits of generators and the SoC of energy storage

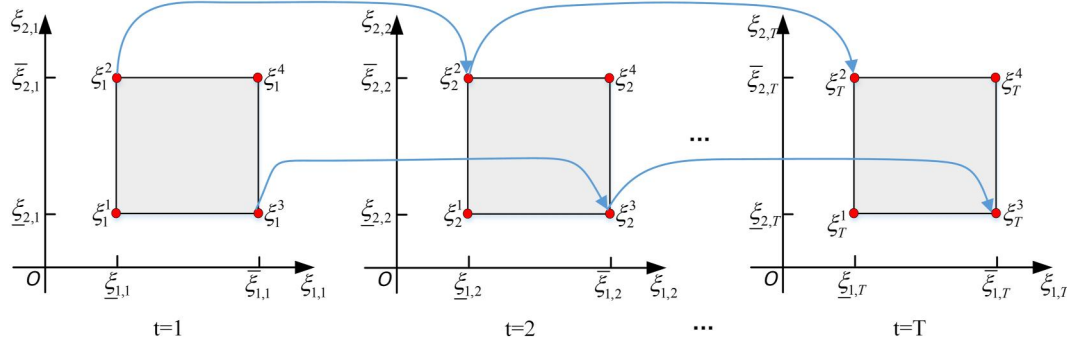


FIGURE 5 Parallel vertex scenario selection

couple different periods. Hence, we introduce first the non-anticipativity constraints of generators and then those of energy storage

- (2A) Non-anticipativity constraints of generator [74]. For any generator i in a power system for $t = 1 : T$, these constraints are

$$b_{i,t} P_i^G \leq P_{i,t}^{G,\min} \leq P_{i,t}^G \leq P_{i,t}^{G,\max} \leq b_{i,t} \bar{P}_i^G, \forall s \in SVS \quad (31a)$$

$$P_{i,t}^{G,\max} - P_{i,t-1}^{G,\min} \leq \Delta_{i,t}^{up} \quad (31b)$$

$$P_{i,t-1}^{G,\max} - P_{i,t}^{G,\min} \leq \Delta_{i,t}^{down} \quad (31c)$$

where $\underline{P}_i^G / \bar{P}_i^G$ is the given physical lower/upper bound of the generation power; $P_{i,t}^{G,\min}$ and $P_{i,t}^{G,\max}$ are specific variables that are determined in day-ahead unit commitment and used in economic dispatch. SVS is the set of selected vertex scenarios. $\Delta_{i,t}^{up} / \Delta_{i,t}^{down}$ denotes the ramping-rate up/down limit from slot $t - 1$ to t , whose full expression is given in (14)–(15) in Ref. [74].

Constraint (31a) is added into the unit commitment problem (24); the obtained $P_{i,t}^{G,\min}$ and $P_{i,t}^{G,\max}$ are passed to the economic dispatch phase. In economic dispatch, the generator is dispatched in each slot t with $P_{i,t}^G \in [P_{i,t}^{G,\min}, P_{i,t}^{G,\max}]$. Consequently, the maximum generation rise between two adjacent slots is $P_{i,t}^{G,\max} - P_{i,t-1}^{G,\min}$, so we add (31b) to make such a rise reachable. Similarly, the maximum generation drop between two adjacent slots is $P_{i,t-1}^{G,\max} - P_{i,t}^{G,\min}$, so we add (31c) to make such a drop reachable.

- (2B-I) Non-anticipativity constraints of energy storage-Type I [65]: this type generates specific bounds for the SoC of energy storage. To this end, consider the following non-anticipativity constraints for storage j in any t :

$$\underline{E}_j \leq E_{j,t}^{\min} \leq E_{j,t,s} \leq E_{j,t}^{\max} \leq \bar{E}_j, \forall s \in SVS \quad (32a)$$

$$-\bar{P}_j^{dis} \leq f(E_{j,t}^{\max} - E_{j,t-1}^{\min}) \leq \bar{P}_j^{ch} \quad (32b)$$

$$-\bar{P}_j^{dis} \leq f(E_{j,t}^{\min} - E_{j,t-1}^{\max}) \leq \bar{P}_j^{ch} \quad (32c)$$

where $\underline{E}_j / \bar{E}_j$ are the given physical lower/upper bounds of SoC and $\bar{P}_j^{dis} / \bar{P}_j^{ch}$ are the given physical upper bounds of discharging/charging power. $f(\cdot)$ is the function that connects the real SoC change of storage to the grid-side power with energy exchange efficiency; please see (39) in Ref. [65].

Constraint (32a) is added to the unit commitment problem (24); the obtained $E_{j,t}^{\min}$ and $E_{j,t}^{\max}$ are passed to the economic dispatch phase. In economic dispatch, energy storage is dispatched in each slot t with $E_{j,t} \in [E_{j,t}^{\min}, E_{j,t}^{\max}]$, and similar situations are discussed: the maximum SoC rise between two adjacent slots is $f(E_{j,t}^{\max} - E_{j,t-1}^{\min})$, so we add (32b) to make such a rise reachable; the maximum generation drop between two adjacent slots is $f(E_{j,t}^{\min} - E_{j,t-1}^{\max})$, so we add (32c) to make such a drop reachable.

- (2B-II) Non-anticipativity constraints of energy storage-Type II [75]: this type generates specific bounds for the charging and discharging power of energy storage. To this end, consider the following non-anticipativity constraints for storage j in any t :

$$0 \leq P_{j,t}^{d,\min} \leq P_{j,t,s}^d \leq P_{j,t}^{d,\max} \leq \bar{P}_j^{dis} b_{j,t}, \forall s \in SVS \quad (33a)$$

$$0 \leq P_{j,t}^{c,\min} \leq P_{j,t,s}^c \leq P_{j,t}^{c,\max} \leq \bar{P}_j^{ch} (1 - b_{j,t}), \forall s \in SVS \quad (33b)$$

$$E_{j,t}^{\min} = E_{j,t-1}^{\min} + P_{j,t}^{c,\min} \eta_j^c - P_{j,t}^{d,\max} \eta_j^d \quad (33c)$$

$$E_{j,t-1}^{\max} = E_{j,t}^{\max} + P_{j,t}^{c,\max} \eta_j^c - P_{j,t}^{d,\min} \eta_j^d \quad (33d)$$

$$\underline{E}_j \leq E_{j,t}^{\min} \leq E_{j,t}^{\max} \leq \bar{E}_j \quad (33e)$$

where $b_{j,t}$ is a binary variable and $b_{j,t} = 1/0$ represents storage j discharges/charges in slot t . η_j^c / η_j^d is charging/discharging

efficiency. $P_{j,t}^{d,\min}$, $P_{j,t}^{d,\max}$, $P_{j,t}^{c,\min}$, $P_{j,t}^{c,\max}$, $E_{j,t}^{\min}$, and $E_{j,t}^{\max}$ are all auxiliary variables.

The unit commitment problem (24) considers constraint (33b); $P_{j,t}^{d,\min}$, $P_{j,t}^{d,\max}$, $P_{j,t}^{c,\min}$, and $P_{j,t}^{c,\max}$ are solved and passed to the economic dispatch phase, and the energy storage is dispatched in each slot t with $P_{j,t}^d \in [P_{j,t}^{d,\min}, P_{j,t}^{d,\max}]$ and $P_{j,t}^c \in [P_{j,t}^{c,\min}, P_{j,t}^{c,\max}]$. With such allowable ranges, the lowest SoC in each slot t is $E_{j,t}^{\min}$, and the highest is $E_{j,t}^{\max}$, as shown in (33c) and (33d), respectively. Then, $E_{j,t}^{\min}/E_{j,t}^{\max}$ are bounded below/upper by $\underline{E}_j/\bar{E}_j$ in (33e).

In the above implicit policy method, two major steps are extracted from the existing literature, and the way to obtain the auxiliary bounds is clarified. For energy storage, it is enough to choose one group of non-anticipativity constraints either in (2B-I) or (2B-II). Some additional remarks are given:

- (1) The generator or energy storage is dispatched in each slot t with those bounds and ranges. The existence of at least one feasible and causal solution to any $\xi_t \in \Xi_t$ is guaranteed by vertex scenarios; the temporal couplings are handled by the ranges defined in non-anticipativity constraints. Please note that the non-anticipativity constraints in implicit policy do not literally guarantee time causality.
- (2) At the beginning of Section 2.3.3, it is mentioned that the implicit policy method is to identify a specific feasible region for dispatch variables, which is finally described by those ranges. They are sufficient conditions for the multistage robust feasibility of economic dispatch while the necessity is not clear [65].
- (3) The explicit policy restricts the redispatch function, and the implicit policy restricts the feasible region of decision variables. Hence, both methods cause conservativeness but there is no general conclusion as to which one is more conservative. In the authors' experience, for an economic objective related to power generation, the explicit policy performs better since the affine correlation properly reflects the adjustment of generation in response to the variation of renewable energy resources; the feasible region obtained by the implicit policy method may exclude some good solutions. However, it will be different if we consider some special objectives, for example, total voltage deviation; the reason is that the deviation from the reference, that is, $|v - v_{\text{ref}}|$, is not a monotonous function of the nodal voltage v , so an affine adjustment rule may lead to unexpected bad performance. But overall, this is a case-by-case issue.

To reduce the conservativeness of the implicit policy, it is helpful to characterise the exact feasible region of dispatch variables in each slot t . Taking SoC as an example, 'exactness' means this region only consists of the SoC levels that can achieve multistage robust feasibility, so $[E_{j,t}^{\min}, E_{j,t}^{\max}]$ in (2B-I) is a subset of it. This implies that selecting a dispatch solution from this exact region is the sufficient and necessary condition

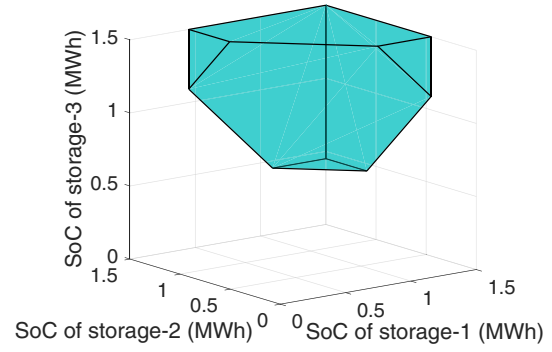


FIGURE 6 Dynamic robust feasibility region [76]

for the multistage robust feasibility. For the economic dispatch problem, a pioneering study in Ref. [76] looks into the multistage robust feasibility operation of a distribution system with renewable generation and energy storages. Specifically, x_t and y_t denote state and non-state variables, and w_t is the uncertainty that rests in a polyhedral uncertainty set $\mathbb{W}_t$; considering a distribution system without local generator, whose power flow is modelled by linearised branch flow, the operating constraint in each slot $t \in \mathbb{T}$ is

$$Ax_t + By_t + Cx_{t-1} \leq b_t + Dw_t, \forall t \in \mathbb{T} \quad (34)$$

where A , B , C , b and D are coefficient matrices. Motivated by the optimality recursion in DP, the so-termed dynamic robust feasible region $\mathcal{F}_t$ is defined by the following feasibility recursion:

$$\mathcal{F}_t = \left\{ x_t \mid \forall w_{t+1} \in \mathbb{W}_{t+1}, \exists x_{t+1} \in \mathcal{F}_{t+1}, y_{t+1} : \begin{array}{l} Ax_{t+1} + By_{t+1} + Cx_t \leq b_{t+1} + Dw_{t+1} \end{array} \right\} \quad (35)$$

The dynamic robust feasibility region defined in (35) implies that if x_t is chosen from $\mathcal{F}_t$, there must be at least one feasible operating strategy (x_{t+1}, y_{t+1}) for any uncertainty realisation $w_{t+1} \in \mathbb{W}_{t+1}$ in the next slot. Then, $\mathcal{F}_t$ is proven to be polyhedral and can be calculated by the polyhedral projection [76]. In Figure 6, the dynamic robust feasibility region in period 18 is given as a polyhedral set for SoC levels of three energy storage units, which are exactly the allowable region of state variables in that problem.

As long as the operating problem is linear, this region makes sense for multistage robust feasibility guarantee. A future direction is to generalise the method to handle binary variables.

3 | ONLINE OPTIMISATION

Revisiting the multistage optimisation methods discussed above, the approximate value functions in Sections 2.1 and 2.2 are computed and stored offline. In the online application, only simple (one stage) optimisation problem is solved. In other words, the calculation of value functions resembles a training or learning process. Differently, online optimisation

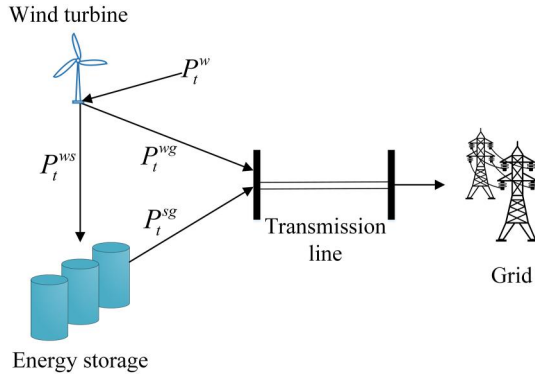


FIGURE 7 Structure of a wind-storage integrated plant

emphasises a pure online decision-making process without referring to historical data or forecast information. Some online optimisation algorithms make decisions while learning how to make decisions better, which can be regarded as a process of online learning [77]. Next, we introduce some typical online optimisation methods for energy storage dispatch and control with renewable integration.

3.1 | Lyapunov optimisation

In control theory, a Lyapunov function is used to certify the stability of a dynamic system. In this section, Lyapunov optimisation refers to a special technique for the online optimal control of stochastic queueing networks. This technique is first developed in the renowned textbook [78] for stochastic network optimisation with application to communication and queueing systems. To better explain how this method is applied in energy storage dispatch and control, we will use a wind-storage integrated plant operation problem for illustration [79]. Considering the system structure in Figure 7, the wind-storage system participates in real-time energy market for arbitrage in a self-scheduling manner; the amount of electricity sold to the grid in each τ -duration slot t is $f_t = \tau(P_t^{wg} + P_t^{sg}\eta_d)$ where η_d is the discharging efficiency of energy storage. Uncertainty stems from the electricity price π_t and wind resources P_t^w . To minimise (maximise) the long-term costs (profits), the deterministic operating model is

$$\begin{aligned}
 \min \quad & \sum_{t=0}^{T-1} -\pi_t f_t \\
 \text{s.t.} \quad & P_t^{ws} \geq 0, P_t^{wg} \geq 0, P_t^{sg} \geq 0, \forall t \\
 & P_t^{wg} + P_t^{ws}/\eta_c \leq P_t^w, \forall t \\
 & P_t^{wg} + P_t^{sg}\eta_d \leq T^u, \forall t \\
 & P_t^{ws} \leq P^u, P_t^{sg} \leq P^u, \forall t \\
 & E_{t+1} = E_t + \tau(P_t^{ws} - P_t^{sg}), \forall t \\
 & E^l \leq E_t \leq E^u, \forall t
 \end{aligned} \tag{36}$$

For brevity, the operating constraints in (36) except the last two are abbreviated to CONS-OPER_t, $\forall t$.

However, solving the deterministic optimisation model requires the entire price and wind generation sequences, which can be done as offline simulation tasks. In real dispatch, the uncertainty is revealed in sequence. Although uncertain wind power can be predicted in some ways, the prediction error may be large, and the resulting dispatch solution may be highly suboptimal. The Lyapunov optimisation paves the way for making instant dispatch decisions in response to the newly observed uncertainty realisation.

To this end, it first needs to deal with the last two constraints in model (36). Given the initial SoC E_0 , summing up both sides of the SoC dynamic equation for $t = 0 : T-1$ and taking expectation over randomness give rise to

$$\mathbb{E}[E_T] - E_0 = \sum_{t=0}^{T-1} \mathbb{E}[\tau(P_t^{ws} - P_t^{sg})] \tag{37}$$

Further, E_T must be bounded; so dividing both sides of Equation (37) by T and taking limits as $T \rightarrow \infty$ produce

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\tau(P_t^{ws} - P_t^{sg})] = 0 \tag{38}$$

Condition (38) manifests that in a long time horizon, the expected total charged power is equal to the total discharged power. Besides, online dispatch intends to guarantee long-term performance by only scheduling the current time slot; a natural objective is to minimise the long-term time-average cost. Eventually, the time-average formulation of the self-dispatch problem can be cast as

$$\begin{aligned}
 \min \quad & \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[-\pi_t f_t] \\
 \text{s.t.} \quad & \text{CONS - OPER}_t, \forall t; \tag{38}
 \end{aligned} \tag{39}$$

Problem (39) is a relaxation of (36) since condition (38) is softer than the last two constraints in (36). However, condition (38) is still time-coupled.

Define a virtual queue $Q_t = E_t - \gamma$ where γ is a constant offset. Obviously, Q_t and E_t share the same dynamics, so condition (38) implies

$$\lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E}[Q_T] = 0 \tag{40}$$

which is to say the virtual queue Q_t is mean rate stable [80]. So we have

$$\begin{aligned}
 \min \quad & \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[-\pi_t f_t] \\
 \text{s.t.} \quad & \text{CONS - OPER}_t, \forall t; \\
 & \text{Queue } [Q_t] \text{ is mean rate stable (40)}
 \end{aligned} \tag{41}$$

The Lyapunov function of Q_t is defined as $L_t = (Q_t)^2 = (E_t - \gamma)^2$, and the Lyapunov drift is $\Delta_t = L_{t+1} - L_t$. Lyapunov optimisation indicates that minimising the Lyapunov drift can maintain the system state (queue) stability. Meanwhile, what the decision-maker cares about is not only the system stability but also the income from selling electricity. To strike a balance between the economic payoff and system stability, the dispatch strategy is obtained by simultaneously minimising an upper bound of the sum of drift and the instant cost, namely the drift-plus-penalty, which is expressed by $\Delta_t + V(-\pi_t f_t)$, giving rise to the following problem

$$\begin{aligned} \min \quad & \tau Q_t (P_t^{ws} - P_t^{sg}) + V(-\pi_t f_t) \\ \text{s.t.} \quad & \text{CONS - OPER}_t \end{aligned} \quad (42)$$

where V is a weight coefficient that compromises two objectives. Problem (42) is time-decoupled; dispatching this wind-storage system via solving (42) in each slot t realises an explicit optimality guarantee, as proven in Ref. [78]. This is the main idea of Lyapunov optimisation.

E_t is inherently a queue; introducing the auxiliary Q_t is to restrict E_t within the stable range $[E^l, E^u]$ with the collaboratively chosen perturbation γ and weight coefficient V . This choice depends on specific problems; in this case, $\gamma = E^l + \tau P_t^U + V\pi_{\max}$ and the allowable range of V is given in Theorem 1 in Ref. [79]. Further, a larger V can improve the performance; the reason is that a larger V means a heavier weight on penalty term; however, if V is too large, the constraint $E^l \leq E_t \leq E^u$ cannot be satisfied because the weight on drift term is too minimal. A valid choice of V is given in Theorem 2 in Ref. [79].

The Lyapunov optimisation with drift-plus-penalty algorithm has been reported in many storage-related scenarios. A typical application is dynamic or real-time energy management, as in Ref. [81] for smart grid with distributed energy resources, in Ref. [82] for microgrids, in Ref. [83] for energy hub, and in Ref. [84] for residential customers. Besides, Lyapunov optimisation is used to balance the power in a general renewable-integrated power grid in Ref. [85] and cooperatively operate the

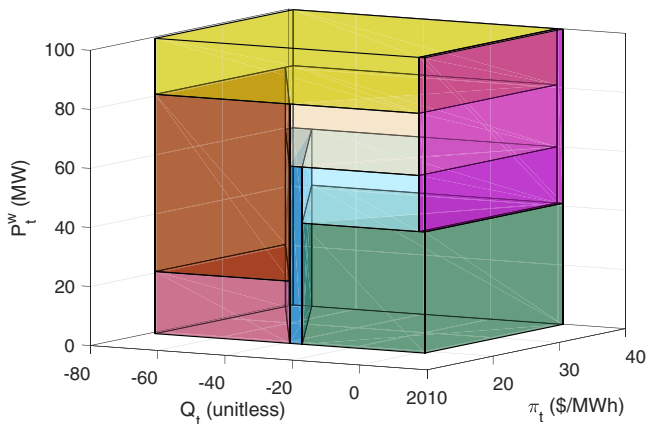


FIGURE 8 Subdivisions of the allowable range of (π_t, P_t^w, Q_t) [79]

storages in a power network in Ref. [86]. The above studies focus on Lyapunov optimisation methods with infinite time horizon since this infinity can facilitate the proof of algorithm performance. Li and Dong explore the finite-time horizon Lyapunov optimisation and verify it in energy storage management [87–89].

Learning from the aforementioned literature, which is based on Lyapunov optimisation, one can conclude the main steps to accommodate such a paradigm. First, build the offline deterministic optimisation model; then, relax the time-coupling constraints of energy storage and reformulate the problem in a time-average form with queue stability conditions; next, develop the drift-plus-penalty online problem with Lyapunov function and drift; finally, give with proof the performance guarantee as well as the choice criterion for queue perturbation and weight coefficient, if necessary. However, it is found that some operating constraints, such as the strict power balance, may jeopardise the effectiveness of Lyapunov optimisation. The reason is that the performance of this method is realised by minimising the drift-plus-penalty, whose gradients with respect to discharging and charging power control the energy storage; if there is a strict equality constraint, it will dominate the decision-making and invalidate the drift-plus-penalty. Therefore, it is important to formulate a proper optimisation model when using Lyapunov optimisation.

Lyapunov optimisation is an effective tool for the online decision-making problem with time-coupling system state such as the SoC of energy storage, thanks to its independence of predictions and explicit performance guarantee. At each stage, a single-period optimisation problem with newly available information is solved, so it is possible to calculate an analytical policy from observations to decisions, which is appealing in practice use. For instance, in Ref. [79], multiparametric programming techniques are employed to extract the analytically dispatch policy, which maps the electricity price π_t , wind resources P_t^w , and the current queue Q_t to the actions to be taken. Such analytical relations are piecewise affine functions,

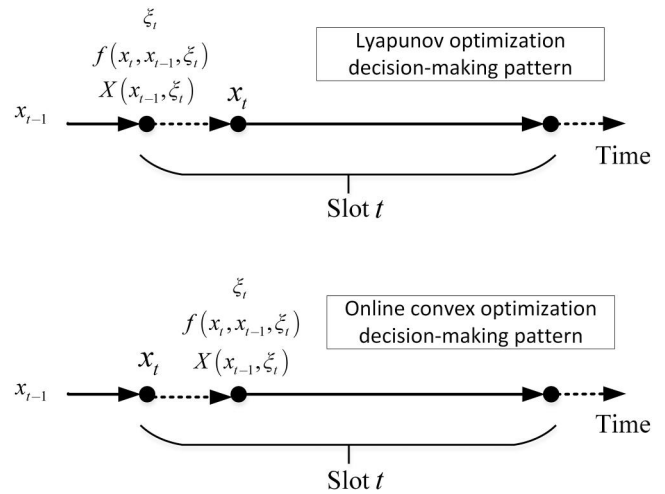


FIGURE 9 Decision-making pattern of Lyapunov optimisation and online convex optimisation

and each piece corresponds to one subdivision of the allowable range of (π_t, P_t^w, Q_t) , as seen in Figure 8 with eight subdivisions in total; to obtain the dispatch action is to find which subdivision the observation of (π_t, P_t^w, Q_t) falls into.

Most of the above studies establish the drift-plus-penalty algorithm following the path of Lyapunov optimisation theory. In fact, the drift-plus-penalty formulation can be deduced from the perspective of stochastic dual gradient approach [90]: the queue Q_t in the objective function can be regarded as the Lagrange multiplier with regard to the state variation constraint in a stationary stochastic programme, for example, problem (9) in Ref. [90]; repeatedly solving the drift-plus-penalty problem can be seen as the stochastic dual gradient iterations. From this perspective, some techniques are developed to improve the algorithm performance; the key idea is to break the direct tie between Lagrange multiplier and SoC level E_t [91]. In Ref. [92], a learn-and-adapt Lagrange multiplier updating rule is proposed, in order to learn from the historical data and then adapt the upcoming decisions. Additional discussions and interpretations on Lyapunov optimisation and its variants are found in Refs. [93, 94].

3.2 | Online convex optimisation

As shown in Figure 9, the Lyapunov optimisation determines x_t with cost function $f(\cdot)$ and feasible region $X(\cdot)$ defined by the observed uncertainty realisation ξ_t . Such a decision-making pattern covers several problems in power systems; for example, the retail electricity price is sent to the energy storage plant, after which the arbitrage action is taken. However, sometimes, at the moment the action is taken, the uncertainty-dependent cost function and feasible region are unavailable and will be revealed later. For example, when the energy storage participates regulation market, it bids a capacity before market clearance. The price and the profit function become clear after the market is cleared [95]. In demand response programs, the load serving entity needs to dynamically determine the price and send it to the consumers as incentives; the outcome of real-time pricing is revealed after the consumers' response [96].

Online convex optimisation (OCO) is designed for this specific decision paradigm, which can be viewed as a repeated game between a learner and nature [97]: in each slot t , the learner selects an action x_t from a fixed convex strategy set $\mathbb{X}$; subsequently, the nature announces a (possibly adversarial) convex cost function $f_t(\cdot)$ and a convex constraint function $g_t(\cdot)$. As a result, the learner receives a cost of $f_t(x_t)$ and a constraint condition $g_t(x_t) \leq 0$.

If all the cost and constraint functions at all T slots are known a priori, the optimal strategy $x^* = (x_1^*, x_2^*, \dots, x_T^*)$ can be obtained by solving the following benchmark problem:

$$\begin{aligned} (x_1^*, x_2^*, \dots, x_T^*) = \arg \min_{x_1, x_2, \dots, x_T} \sum_{t=1}^T f_t(x_t) \\ \text{s.t. } g_t(x_t) \leq 0, t = 1 : T \end{aligned} \quad (43)$$

With x^* , the T -slot process achieves the global optimality while maintaining all the constraints, which however is difficult to implement online, as functions $f_t(\cdot)$ and $g_t(\cdot)$ are unknown before the decision x_t is taken in each period t . In view of this, two measures are used to define the performance of OCO algorithms [98]; one is the regret, which is the difference between the cost incurred by the OCO algorithm and the hindsight optimum

$$\text{Reg}^d = \sum_{t=1}^T [f_t(x_t) - f_t(x_t^*)] \quad (44)$$

where $x_t, t = 1 : T$ is one decision sequence provided by the algorithm under investigation. The regret quantifies the sub-optimality caused by the online policy.

The other measure is about the constraint violation; because the decision x_t is taken before the constraint function $g_t(\cdot)$ is announced, there may be a constraint violation; thus, the accumulative constraint violation is the other measure, that is,

$$\text{Vio} = \left\| \left[\sum_{t=1}^T g_t(x_t) \right]_+ \right\| \quad (45)$$

In some studies, for example, Ref. [99, 100], the baseline strategy remains constant throughout the horizon, which is obtained in hindsight by solving

$$\begin{aligned} x_s^* = \arg \min_{x_s} \sum_{t=1}^T f_t(x_s) \\ \text{s.t. } g_t(x_s) \leq 0, t = 1 : T \end{aligned} \quad (46)$$

Therefore, the dynamic regret in (44) seems more reasonable than a static one in power system operation problems. The definition of accumulative constraint violation in (45) is independent of any baseline strategy.

In addition to how the regret is defined, another criterion that distinguishes different OCO algorithms is what information is revealed to the learner after each decision x_t is taken. If the analytical expressions of functions $f_t(\cdot)$ and $g_t(\cdot)$ or their gradients are available to the decision-maker, as we assume above, this is the OCO with full feedback; if only the function values at one or multiple points are available, this is the OCO with bandit feedback [101]. The main handicap to solving OCO with bandit feedback is the lack of gradients of $f_t(\cdot)$ and $g_t(\cdot)$ at x_t , which is very important in a convex optimisation algorithm. Consequently, a key step in bandit OCO is to

TABLE 2 Online convex optimisation studies for different feedback and regret

Feedback	Regret	
	Static	Dynamic
Full	[97, 99, 100, 102]	[103–105]
Bandit	[106–108]	[98, 109, 110]

estimate the gradient using limited information. Table 2 summarises the representative research papers of OCO with different feedback and regret definitions.

OCO algorithm is developed to achieve a low regret and a small constraint violation. Basically, all the existing algorithms are online extensions or variants of the classical Arrow–Hurwicz saddle point method [111]. Considering the cost function $f_t(\cdot)$ and constraint condition $g_t(\cdot) \leq 0$, the Lagrangian in slot t is

$$\mathcal{L}_t(x_t, \lambda_t) = f_t(x_t) + \lambda_t^\top g_t(x_t) \quad (47)$$

where λ_t is the column vector of Lagrange multipliers. The saddle point algorithm takes successive primal gradient descent steps and dual gradient ascent steps [112], that is,

$$\begin{aligned} x_{t+1} &= \mathcal{P}_{\mathbb{X}}[x_t - \eta \nabla_x \mathcal{L}_t(x_t, \lambda_t)] \\ \lambda_{t+1} &= \mathcal{P}_{\Lambda}[\lambda_t + \eta \nabla_{\lambda} \mathcal{L}_t(x_t, \lambda_t)] \end{aligned} \quad (48)$$

where $\Lambda = \{\lambda | \lambda \geq 0\}$ is the feasible region of Lagrange multipliers; η is the step-size parameter; $\mathcal{P}_{\mathbb{S}}[s]$ denotes the orthogonal projection of the given point s onto the convex compact set $\mathbb{S}$, that is the point in this set shortest to s .

To improve the performance, the modified online saddle point method is proposed in Ref. [103]; one modification is to update the primal and dual variables in a Gauss–Seidel manner, instead of the Jacobi manner in (48); the other is to use the exact constraint function instead of its first-order approximation in the primal update. Using the modified Lagrangian for better performance is also reported, as in Ref. [98] where a regularisation term is added to prevent the dual variables from becoming excessively large; such a modification is applied in distributed OCO with gradient feedback [104] and bandit feedback [109]. Comparisons of different OCO algorithms on their performance are given in Refs. [104, 109].

OCO has been applied in proactive network resource allocation [103] and internet-of-things management [113]. Specifically, in the field of power systems, OCO is adopted for pricing or participating demand response in real time [96, 114]. In Ref. [115], the problem that a multi-energy building with energy storage provides ancillary services to the grid is solved by OCO. The distributed control of battery energy storage for frequency regulation is investigated in Ref. [116]; the OCO framework is justified to be more effective than those prediction-based algorithms. This method also makes sense in the distributed charging control of electric vehicles [117]. Using OCO to control battery energy storage for improved frequency regulation is reported in Ref. [118].

Currently, the application of OCO in power systems and energy storage is rather limited. The main reason is that it requires a less common decision paradigm in which the objective and constraint information is perceived as feedback after the decisions are taken. Nonetheless, with the evolution of power systems, the OCO method shows a great potential for the implementation in future research, because it (1) does not need any prediction information for online decision-making and is easy to implement, (2) has clear performance

guarantee, (3) can be executed in a distributed manner, and (4) naturally has a privacy-preserving capability. In particular, the last trait will be further discussed in Section 5 to suggest some future directions.

3.3 | Receding-horizon control and its variants

Lyapunov optimisation and OCO have one thing in common that they are prediction-free. In practice, there might be some relatively exact predictions on the near future data. If the predictions can be updated frequently, receding-horizon control (RHC), also known as model predictive control, is a popular method that utilises such information to make look-ahead decisions.

Generally speaking, the RHC algorithm considers a prediction window w of uncertainty (disturbances) and optimises this horizon subject to system constraints; the optimiser in the nearest period is deployed; then, the algorithm proceeds in a rolling-horizon manner [119]. RHC has a very long history of

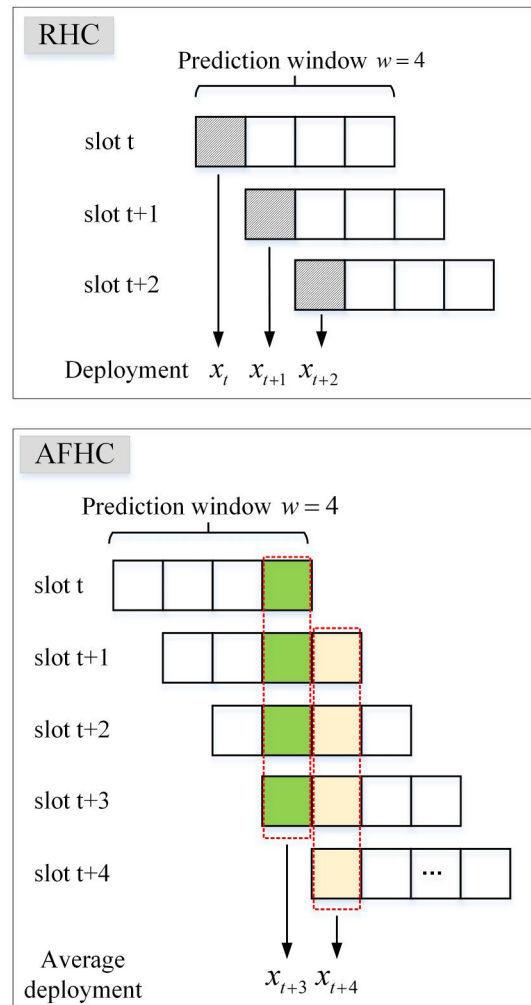


FIGURE 10 Graphical illustration of receding-horizon control and average fixed horizon control

research and is an easy-to-use tool in application. In Ref. [120], in order to maximise the payoff of a wind-storage plant from selling electricity to the main grid, an RHC-based coordinated scheduling framework is presented, which finally resorts to solving a quadratic programming problem to compute the operating strategy. RHC is also used in Refs. [121, 122] to design a controller to smooth the wind power output and improve the wind farm dispatchability with battery storage. Besides, in Ref. [123], RHC serves to control the energy flows in a distribution network with energy storages and variable renewable generations, to make the power flow at a node level track the day-ahead profile.

RHC deploys the solution only in the nearest period. Recently, some work highlighted that this method may result in unexpected bad performance in specific scenarios and thus the performance may not be robust [124]. Unsurprisingly, the root cause is that the entire future information is never perfectly available to RHC, and the optimal strategy may be sensitive to these parameters. To address this limitation, several variants of RHC are proposed with the main idea of taking full advantage of the predictions and reducing the negative impact of forecast errors. In Ref. [124], the average fixed horizon control (AFHC) algorithm is designed, which works by taking the average of many fixed horizon control algorithms. A graphical illustration of RHC and AFHC is given in Figure 10. As the prediction window is $w = 4$, we have the strategy in stage $t + 3$ in periods $t \sim t + 3$. Hence x_{t+3} is the average of the current and all previous values.

Furthermore, the committed horizon control, which generalises both RHC and AFHC, is introduced in Ref. [125]; the commitment level is v , implying v of w steps are averaged to obtain a policy. A characterisation is provided in Ref. [125] that describes when AFHC outperforms RHC and vice versa as well as when committed horizon control policies outperform both RHC and AFHC. The synchronised fixed horizon control is given in Ref. [126], which is reported to be able to achieve a constant, dimension-free competitive ratio even in the non-convex setting given no predictions, and leverage predictions to achieve a competitive ratio that converges to 1 as the prediction window approaches infinity.

To conclude, if some high-quality predictions are available in online decision-making process, RHC and its variants are good choices. A final note is the decisions provided by these methods are definitely non-anticipative, since it is the uncertainty predictions rather than realisations that are utilised in decision-making.

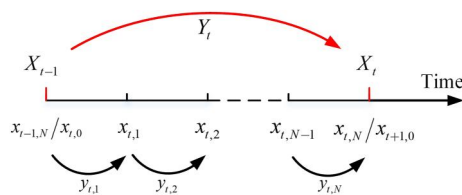


FIGURE 11 Two timescale decision-making

For these three online optimisation methods, the former two entail solving single-period optimisation problems at any period while RHC needs to solve a look-ahead optimisation problem, which involves multiple periods. Therefore, RHC is computationally more expensive; the solution time is really a concern, especially for large-scale problems with high time resolution.

4 | MULTI-TIMESCALE OPTIMISATION

As previously mentioned, a series of application scenarios of energy storage involves multiple timescales, so the multi-timescale problem has become a very important topic recently. However, the physical interdependence across timescale makes the whole problem difficult to manage, and the uncertainty also challenges the modelling and solution process, arising the need for systematic multi-timescale optimisation methods.

4.1 | A vertical framework in power system operation

The operating mechanism of power systems inspires this vertical framework of multi-timescale optimisation. For instance, the system operator determines the slowest decisions in day-ahead stage, such as the unit commitment solution [127]; then, those regulation devices with moderate responsive capacity are dispatched to mitigate the hourly uncertainty; in the real-time stage, the most flexible resources, such as energy storage, are dynamically scheduled in response to very rapid fluctuations and real-time power mismatch.

Under such a framework, the multi-timescale active and reactive power coordinated optimal dispatch in active distribution networks is investigated in Ref. [128] considering multiple correlations of renewable energy resources. First, the day-ahead stage minimises the expected daily operation costs and determines the hourly state of on-load tap changers and capacitor banks; then, every 4 h, the intraday stage sets an operating baseline for micro gas-turbine and energy storage, which is further optimised in 30-min intervals. In this process, each optimisation problem is formulated as a scenario-based stochastic programme. In Ref. [129], a temporal-spatial reserve coordination strategy for active distribution networks is proposed, in order to fully exploit multiple resources to reduce the impact of uncertainty on both distribution and transmission networks. Similarly, decisions are taken in three different timescales: the operation and reserve allocation of transmission network in day-ahead stage, the output baseline of distributed energy resources in distribution network in intraday stage, and the deviation accommodation in the real-time stage. Besides, a hybrid timescale dispatch hierarchy for combined heat and power system is proposed in Ref. [130], which is expected to coordinate day-ahead, intraday, and real-time dispatches, as well as the automatic generation control.

Generally speaking, this framework is proper for power system operation with many regulation devices, whose different responsive characteristics entitle them to be dispatched or controlled in multiple timescales with multiple uncertainties [131]. The optimisation problems of coarse timescales usually follow a stochastic or robust paradigm [132]; the fine timescale makes sequential decisions by RHC or other multistage methods. However, this vertical framework considers few details of the finer timescale when making decisions in the coarse timescale, and hence may be suboptimal or even cause infeasibility in the fast timescale operation.

4.2 | Multi-timescale dynamic programming

If we are considering a single-timescale decision-making process, a classical method is used to formulate it as a multistage optimisation problem; DP is employed to solve this problem, as mentioned in Sections 2.1 and 2.2. It is a natural idea to extend DP into those situations with multiple timescales, giving rise to multi-timescale DP.

Considering an example of two-timescale DP, sequential slow and fast decisions are illustrated in Figure 11; at the beginning of each coarse time interval t , the slow action Y_t is taken and the slow system state changes from X_{t-1} towards X_t ; then, at each fine time interval t, n , an action $y_{t,n}$ is selected and consequently the fast system state changes from $x_{t,n-1}$ to $x_{t,n}$.

The modelling of such a two-timescale problem using DP is rather complicated; it depends on specific scenarios. If the slow and fast system states have different physical meanings, for example, considering the SoC of two separate energy storages, the state transition function is decoupled. Otherwise, considering a single energy storage in slow energy arbitrage and fast frequency regulation [133], both X_t and $x_{t,n}$ represent the SoC of this energy storage, so we have $X_{t-1} = x_{t,0}$ and $X_t = x_{t,N}$; meanwhile, the evolution of X_t is determined by both the slow action and the fast actions. Another interaction between these two timescales is that the slow action Y_t partially defines the feasible region of fast action $y_{t,n}$. Again, in the above example, one of the slow actions is the fraction of power capacity that is allocated for frequency regulation in the coming coarse time interval; then, this capacity defines the allowable range of fast actions, that is, the frequency regulation responses.

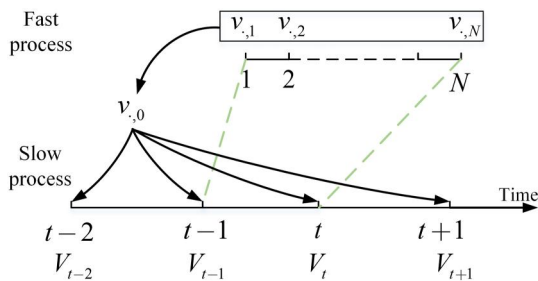


FIGURE 12 Decomposition of two timescale

Ref. [133] is a representative work of multi-timescale DP, which is used in co-optimising a battery storage for joint energy arbitrage and frequency regulation. The arbitrage power is determined every 5 min, and the frequency regulation acts every 2 s; different timescales are coordinated in a nested structure via mutually connected value functions. A valuable technique is that unknown data can be characterised as random variables. For example, at the moment of determining the arbitrage power, the SoC change induced by frequency regulation in the coming coarse time interval is unknown; so it is regarded as a random variable. However, the computation of all the value functions is extremely expensive. It is estimated that at least 60 TB of disk space is needed to store the value functions (in the form of discrete points) in a 1-day horizon. To address this challenge, the twin study in Ref. [134] computes the sparse low-rank approximation of value functions. In Ref. [135], the optimal energy management of wind-battery hybrid power system is formulated as a two-scale DP problem; instead of describing uncertainty in a stochastic or robust manner, this method is simply prediction-based: the long-term ahead prediction of electricity price and wind energy is used to build a deterministic macro-scale DP, which gives the reference trajectory of battery SoC; the micro-scale DP based on 3-h ahead prediction further dispatches the system. Two timescales connect via terminal value functions.

Multi-timescale DP seems elegant since it uses the nested structure to coordinate multiple timescales by value functions. However, two main challenges remain to be addressed (Figure 12):

- (1) The computational burden should be considerably reduced. Even a DP problem with one single timescale consumes numerous efforts in computation, let alone the multi-timescale DP. Timescale decomposition could be efficacious [136], as depicted in Figure 12, if all the fast processes are consistent under identical exogenous conditions.
- (2) The system state transition calculated in different timescales should be coordinated. For example, in the joint energy arbitrage and frequency regulation of energy storage, both X_t and $x_{t,N}$ in Figure 11 physically refer to the SoC of storage. At the coarse timescale, the optimiser X_t^* is obtained considering the instantaneous and future costs, implying that the terminal state is expected to be X_t^* . However, on the one hand, it is hard to estimate the impact of the next fast actions on X_t^* ; on the other hand, the fast actions should be carefully chosen to achieve $x_{t,N} = X_t^*$ or $x_{t,N}$ as close to X_t^* as possible.

4.3 | Multi-timescale extensions of Lyapunov optimisation

In Section 3.2, we introduce Lyapunov optimisation; generalising it into those problems with more than one time granularity suggests multi-timescale Lyapunov optimisation.

The two-timescale Lyapunov optimisation is reported in Ref. [137] in a cost-throughput tradeoff problem in geodistributed data analytics. Without attaining the knowledge of future information, the input data placement and the number of requests to be served are determined in two different timescales. To this end, a Lyapunov drift that spans several slots is defined. To find an upper bound of the drift-plus-penalty problem, the queue backlog in the near future is approximated as the current value, that is, the value at the beginning of each coarse time interval. In Ref. [138], the two-timescale Lyapunov optimisation method is used for predictive service placement; the algorithm here incorporates the prediction of user mobility in the near future. The real-time operation of heterogeneous energy storage units is investigated in Ref. [139]; two battery units collaborate to support a microgrid in different timescales: the discharging power of slower battery exchange and the inflow power from the main grid serve the basic demand; the difference between real-time renewable generation and demand is filled by the fast battery as well as the energy brought from the real-time market. The slower problem involves the expected cost of the fast process, which is solved via a projected subgradient scheme.

A common difficulty to implement the two-timescale Lyapunov optimisation framework is how to consider the stochastic fast process and its impact on slow decisions. Because a period in the slow timescale covers multiple periods in the fast timescale, using stochastic approximations or predictions might be inevitable in the slow decision process, which seems discouraging. This is exactly the challenge in multi-timescale problems.

4.4 | Other representative methods and studies

There are many other multi-timescale optimisation problems in power systems that have also been reported in the literature. We discuss the sources of multi-timescale characteristics and the solutions to them.

Multi-timescale devices: to control fast storage devices in support of power system operation with slow conventional generators, a two-scale stochastic optimisation approach is proposed in Ref. [140]. The generation level of slow-ramping generators is determined at the coarse timescale. The cost that incurs at the fine timescale is due to the usage of storage devices, generation from operating reserves, and transactions with other grids. Unfortunately, this approach considers only one coarse period and makes less sense.

Multi-timescale electricity market: to help a stationary battery system participate in multiple energy markets with different clearing frequencies, a multi-timescale decision-making framework is proposed in Ref. [141], which combines multistage stochastic programming with model predictive control. The slow process is formulated as SDDP and the fast process goes in a rolling-horizon manner. However, an assumption that simplifies this problem is that when the realisation of slow uncertainty is revealed at each coarse time

interval, the fast uncertainty trajectory within this interval is also observed, which is a very strong condition in practice.

Multi-timescale system operation: to address the two-timescale joint economic dispatch and frequency regulation problem in power systems, an elegant framework is proposed in Ref. [142], which decomposes the joint problem without loss of optimality into slow and fast timescale subproblems. The fast subproblems (frequency regulation) are solved using a distributed control algorithm, which preserves network stability during transients. Similarly, only one economic dispatch period is considered.

It can be seen that most existing multi-timescale optimisation methods are customised for applications with specific assumptions or simplifications. The difficulty in multi-timescale problem is the coordination between different timescales; one timescale means a multistage decision-making problem, but a multi-timescale problem cannot be solved by simply using single-timescale optimisation methods for multiple times. New methods remain to be further explored. One possibility is to apply explicit policies in some timescales and thus circumvent sophisticated architecture, at the cost of some optimality.

5 | FUTURE DIRECTIONS

In the coming few decades, the penetration of renewable energy will further grow to achieve carbon neutrality. The role of energy storage is becoming increasingly important. Therefore, we believe energy storage will remain a highly prominent research topic in the foreseeable future. Some interesting directions along the line of energy storage dispatch and control with renewable integration are envisioned.

Single-timescale scenarios:

- (1) The long-term operation of seasonal energy storage [143]. At present, researchers pay more attention to the short-term fluctuation of renewable energy; however, as the penetration level of REG continuously grows, the long-term/seasonal variation cannot be neglected, so seasonal energy storage is of great significance for long-term power balance [144]. Consequently, seasonal energy storage management should be a concern and investigated thoroughly. A challenge is that the long-term uncertainty is more difficult to predict; so the seasonal power system operation should be robust against such randomness.
- (2) Online market participation of commercial storage plant. In the current research, the strategic bidding of energy storage in the power market is typically modelled as a bilevel optimisation problem. As in Ref. [145], the storage system makes strategic offering decisions in the upper level while anticipating the outcome of market clearance in the lower level. The optimal bidding strategy can be obtained via solving an equivalent single-level mathematical programme by replacing the lower level problem with its optimality conditions. Such a framework is instructive but has two main barriers in practice. First, decisions are non-causal

since this model does not satisfy non-anticipativity; in particular, the real-time power market must run sequentially with time-varying conditions. Second, this model assumes that the energy storage owner has all the market information, such as the detailed model of the power grid as well as other market participants' bidding strategies. In real power market operations, any commercial storage unit has no access to power grid data or rivals' bidding strategies. So online learning/optimisation would be a promising approach for this problem. The storage unit submits its bids to the market and then receives only necessary (maybe full or bandit) information from the market once it is cleared, successively learning to optimise bidding strategy and maximise payoffs. Also, this idea can be generalised to similar problems to find a decision sequence in uncertain conditions with partially available information. However, current OCO algorithms have difficulty in considering complex constraints, which deserves further research.

Multi-timescale scenarios:

- (1) Energy storage for joint energy arbitrage and frequency support. In future power systems, most of the conventional generators are replaced by REG and energy storage, so the energy storage unit will play an important role in both frequency regulation and peak shaving. In the current industrial practice, such as in PJM market [146], AGC signal is sent every 2 s, and the real-time energy and ancillary service market are cleared every 5 min; so this is obviously a multi-timescale problem. There should be a proper coordination between the slow capacity allocation and the fast signal response [133].
- (2) Coordination of seasonal and short-term energy storage. With the growing penetration of REG, the seasonal fluctuation of renewable sources is no longer negligible, and thus the seasonal energy storage becomes more indispensable. In general, seasonal energy storage, such as hydrogen storage [147], is large in scale but low in energy conversion efficiency; short-term energy storage, such as electrochemical battery, dominates in response time and efficiency but suffers from limited capacity and cycle life. Therefore, the coordination of seasonal and short-term energy storage is important and challenging under uncertain conditions.

Last but not least, the degradation of battery storage must be considered in the long-term operation. It makes sense to ignore the degradation effect in a short-term operation problem; however, the accumulative degradation is visible with a long enough horizon. The degradation is usually calculated via rainflow counting algorithm by counting the number of cycles as well as the depth of discharging [6] and can be taken into account in two ways: first, adding the degradation cost into the objective to be optimised [148]; second, modelling the loss of useable energy capacity [149]. In a sense, one important task of battery storage management is to manage the degradation.

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CONFLICT OF INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

NOMENCLATURE

ADP	Approximate dynamic programming.
AFHC	Average fixed horizon control.
AGC	Automatic generation control.
DP	Dynamic programming.
MPEC	Mathematical programme with equilibrium constraints.
OCO	Online convex optimisation.
RDDP	Robust dual dynamic programming.
REG	Renewable energy generation.
RHC	Receding-horizon control.
SDDP	Stochastic dual dynamic programming.
SoC	State of charge.
VFA	Value function approximation.

DATA AVAILABILITY STATEMENT

No data.

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